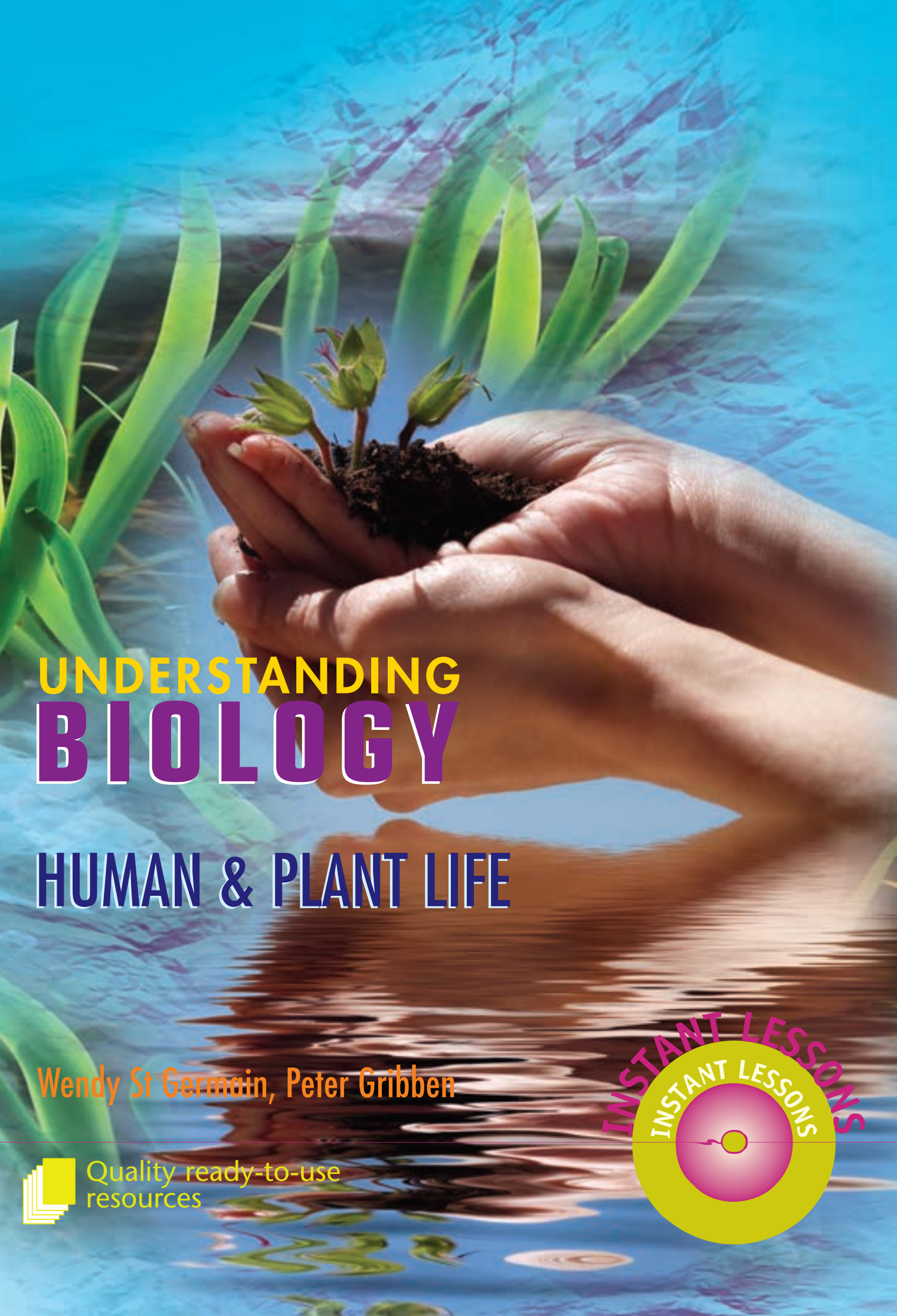




UNDERSTANDING BIOLOGY

HUMAN & PLANT LIFE

Wendy St Germain, Peter Gribben



Quality ready-to-use
resources





UNDERSTANDING **BIOLOGY** HUMAN & PLANT LIFE

Wendy St Germain, Peter Gribben

This series was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover, Final Page Editing and Design provided page design and typesetting services and David Dickson supplied the illustrations.

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Instant Lessons

An imprint of Blake Education Pty Ltd

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ISBN 978 1 92561 097 0



Year 10–12 Biology teachers often report that their students lack basic knowledge of key scientific ideas, which hampers student progress in this subject. The authors of this series have developed activities for topics that, from their experience, they believe are poorly understood by students, as well as for concepts that are particularly crucial to an appreciation of the importance of modern scientific knowledge such as biomaterials, biotechnology and cloning.

In some units, students are required to investigate a topic and provide opinions, and these will make ideal homework or assignment work. Other units are fully self-contained and will test students' understanding of key scientific terms and how this language is used in context.

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The heart pumps blood around the body and is one of the first organs to function in an embryo.

1 What is an *embryo*? _____

The heart circulates blood through the body. The vessels carrying the blood throughout the body are capillaries (narrow), veins (medium-sized) and arteries (large). A human's heart is roughly the size of their fist; for example, a five-year-old child's heart is roughly the size of that child's fist. The heart is made of muscle tissue called *myocardium* (*myo*, meaning 'muscle'). The heart is powered by the sinoatrial (SA) node. The SA node gives a small charge to the tissue, setting the pace of the heartbeat.

Wrapped around the heart is a protective covering, the *pericardium* (*peri*, meaning 'around'). This is a sac-like membrane. The pericardium acts like a special 'cling wrap' around the myocardium.

The heart is divided into four chambers (compartments), two upper and two lower. The left and right sides of the heart are divided by a septum (*septum*, meaning 'enclosure' or 'fence') which acts like a wall between the two halves. Each chamber of the heart is either an *atrium* (plural *atria*) or a *ventricle*.

2 Think of the 'central' part of a house as the area where people first arrive. *Atrium* is an old-fashioned word used to describe the central court of ancient Roman homes. Therefore, for the heart, would the atrium/atria refer to the top chambers or the bottom chambers? Explain your answer. Figure 1 will help.

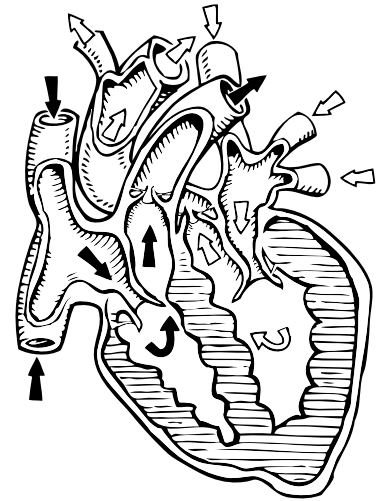


Figure 1

3 Which is the correct order for blood vessels, from largest to smallest?

- a** veins, arteries, capillaries **b** arteries, veins, capillaries **c** capillaries, arteries, veins

The heart deals with both 'fresh' oxygenated and 'stale' deoxygenated (oxygen-poor) blood at the same time. The left atrium sends sufficient fresh blood to the brain and the rest of the body, while the right atrium sends the deoxygenated blood to the lungs, where carbon dioxide is exhaled (and fresh oxygen inhaled). Blood is moved with each heartbeat. In general, arteries carry freshly oxygenated blood from the heart to the head and the rest of the body. Veins carry deoxygenated blood from the body back to the lungs and other 'waste disposal' organs.

4 Name an organ other than the lungs that helps the body to rid itself of waste. _____

5 In biology, a *tissue* is described as 'a collection of cells with a similar composition and function'. Is blood a tissue? Explain.

Confusion sometimes occurs as to which is left and which is right on heart illustrations. They will show the left atrium at the right of the picture! When you look at an illustration, think of where the left and right sides would be *on the organism whose heart is illustrated*. The chambers are named according to the left and right of the individual being illustrated, *not* the reader's left and right.

6 Mark the following statements as true or false. Then correct the false statements so that they are true.

- a** The pericardium forms a special protective 'shield' around the heart. _____
- b** Fresh, oxygenated blood and stale, deoxygenated blood mix in the heart. _____
- c** There are two chambers in the human heart. _____
- d** The sinoatrial node (SA node) sets the pace of the heart's beats. _____
- e** The heart is the main organ of the circulatory system. _____
- f** The upper chambers of the heart are called ventricles. _____
- g** Heart muscle is called myocardium. _____
- h** The lower chambers of the heart are called atria. _____



AV valves

Throughout the blood vascular system, valves direct the flow of blood. Heart valves, (atrioventricular, or AV) are found between the atria and the ventricles. Valves ensure the one-way flow of blood through the heart. As blood leaves the atrium to enter the ventricle, the AV valves open. When the ventricle is full of blood it contracts, forcing the AV valves to close (causing the 'lubb' sound). Once the ventricle has emptied, it 'springs back' to its original shape, (causing the 'dub' sound). Many of the names of valves suggest their function and location.

- 7 What do you think the valves would be called that prevent blood from flowing back into the heart from:
- a the aorta? _____
 - b the pulmonary artery? _____

Sinoatrial node: A closer look at the heart's pacemaker

There is a small cluster of specialised cells in the heart tissue known as the *sinoatrial node*, or pacemaker. The pacemaker produces electrical signals that stimulate cardiac muscles to contract and set the pace of the heartbeats. The pacemaker is located near the right atrium and superior vena cava (the vena cava are major veins – *superior* is at the front or anterior of the heart, the *inferior* is posterior or at the back). An electrical impulse is sent to the atria, where it ripples down to the ventricles. Heartbeat is a result of a steady flow of electrical impulses emitted by the pacemaker. The action of cardiac muscles can be detected as the familiar 'lubb-dub' sound of heartbeat. With every beat, the heart must draw blood into it. The blood is then pushed through the chambers by compressing the muscles, squeezing it out with sufficient force to send it through the body.

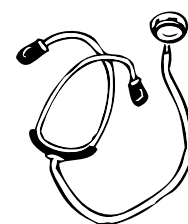


Figure 2

- 8 Name a piece of apparatus used to listen to the heart. _____

Tiny valves open and shut as blood is pushed through the blood vascular system. (A vascular system is a system of fluid-filled spaces.) In doing so, the valves ensure that the flow of blood goes only one way. Blood that has been freshly oxygenated is sent from the heart to the brain and throughout the rest of the body. The freshest blood must go to the head and heart because these organs cannot survive long without it. Special coronary arteries divert fresh blood into the heart muscle as soon as it arrives. Reducing the oxygen flow to the head can lead to brain damage or death. Within 3–5 minutes, brain tissues begin to die. By 10 minutes, less than 1% of victims survive. Those who do survive almost always suffer irreparable brain damage.

- 9 What does *irreparable* mean? _____
- 10 Why do the heart and brain get top priority when it comes to delivery of fresh blood?

- 11 Would someone whose brain was deprived of oxygen for 12 minutes survive? Explain.

Atria and ventricles

Most of the blood enters the heart when it is relaxing, between beats. Blood enters through atria where it then flows downward into the more muscular ventricles. The atria have thinner walls than the ventricles because the blood they receive is largely drawn in through a suction action. Blood then flows in a downward direction to the ventricles with the help of gravity. Ventricles have much thicker walls because they must push the blood they received from the atria in an upward and outward direction.

Blood entering the left atrium is freshly oxygenated, so each red blood cell is carrying an oxygen molecule (O_2). Blood leaves the left atrium, drops into the left ventricle and is sent out, via the aorta, to travel through the body.

Blood entering the right atrium is deoxygenated. This means it has released its oxygen molecule (during its travels) and picked up a carbon dioxide molecule (CO_2) along the way. The deoxygenated blood needs to release its carbon dioxide, and then it can collect more oxygen. It does this when the right ventricle sends it out of the heart, via the pulmonary artery, to the lungs. Once in the lung system, the blood cells release their carbon dioxide (when you exhale) and collect fresh oxygen (when you inhale).

- 12 Why do ventricles have thicker walls than atria? _____



- 13** Write the formula for a carbon dioxide molecule. _____
- 14** Define *vascular*. _____
- 15** Answer the questions below to find the boxed letters. Use the boxed letters to answer part h.
- a** Action caused by electrical impulses to cardiac muscle. [] _____
 - b** Special node that sets the heart's pace. [] _____
 - c** Blood rich in oxygen is said to be [] _____.
 - d** One of the first organs to function in an embryo. [] _____
 - e** The septum acts as a dividing fence or [] _____.
 - f** Thinner walled chamber of the heart. [] _____
 - g** Force of nature that causes blood to flow downward. _____ []
 - h** Use the boxed letters to find this word: I am the blood's _____.
 - i** I am also called the _____ system.

Valves – Valves are tiny structures, made of flaps of tissue that direct the flow of blood through the body (*valva*, meaning 'leaf' or 'door'). Valves are designed to open as the blood flows through them. In the event that any blood flows backward, the pressure of backflow closes the valve. Valves should only 'swing open' in one direction. Think of the effect of backflow as pushing against a 'pull' door. A simplified diagram of a valve in a vein is shown in figure 3.

- 16** Label the valve in figure 3 and show the direction of blood flow.

The heart valves work in a similar manner to those in veins, but are larger.

- 17** Why are the valves in the heart bigger than those in veins?



Figure 3

The opening and closing of valves produce the sound we hear as heartbeat. The heart's contractions ensure blood is sent out from the heart. Blood flowing in the wrong direction, sometimes called 'backwash', would mix freshly oxygenated blood with oxygen-poor blood if the valves did not shut properly.

- 18** Backwash reduces the amount of blood flowing throughout the body. Suggest how. _____
- 19** List a non-body example (e.g. machinery) of when a valve might be used. _____

Naming valves and other bits – easy when you know how!

The word *cusp* describes any kind of crescent-like projection. *Bi-* (2) and *tri-* (3) are prefixes indicating amounts. Hence, bicuspid valves have two crescent-like flaps ('gates') and tricuspid valves have three. The bicuspid valves are found in the left side of the heart, between the atrium and ventricle. As freshly oxygenated blood arrives from the lungs to the *left atrium*, the *bicuspid valves* open, sending blood down to the *left ventricle*.

As the left ventricles contract, the *semilunar valve* opens and blood is delivered to the *aorta*. The aorta is the largest of the arteries in the human body. Semilunar valves, named because of their half moon shape, (*semi*, meaning 'half'; *lunar*, meaning 'moon') prevent backflow from the blood leaving the heart. *Tricuspid valves* are found in the right side of the heart, also dividing the atrium from the ventricle. As the oxygen-poor blood is delivered to the *right atrium*, the tricuspid valves open and send the blood to the *right ventricle*. A set of semilunar valves open, sending the blood to the lungs (via the *pulmonary artery*).

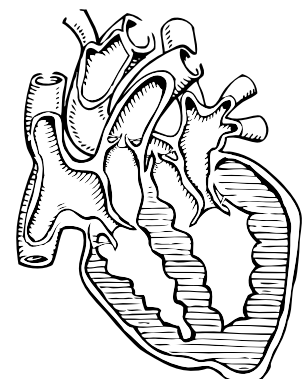


Figure 4

- 20** Label figure 4 using the terms in *italics* in the passage above.
- 21** Show the direction of flow of blood in figure 4.



Division of the heart, by a septum, allows for well-organised movement of blood and produces the effect of two separate pumps. From this arrangement, two circulation paths have evolved. These paths are called the *systemic* and *pulmonary* circuits.

Systemic circuit

Once the exchange of carbon dioxide for oxygen has occurred (at the tiny capillaries of the lungs) the flow of blood progresses to the four pulmonary veins (one pair per lung). All freshly oxygenated blood begins its journey through the circulatory system at the pulmonary veins, which empty freshly oxygenated blood into the left side of the heart. This blood flows to the left atrium and into the left ventricle, which sends it through the aorta to the rest of the body.

- 1 Why is the cardiac tissue forming the ventricles thicker than that of the atria? _____
- 2 Although the ventricles are thicker than the atria (overall), the muscles on the left side of the heart are often thicker than those on the right side of the heart. Suggest why. _____

Pulmonary circuit

The right side of the heart contains oxygen-poor blood on its way to the lungs. Fully circulated blood (carbon-dioxide rich, oxygen poor) travels via the inferior vena cava and superior vena cava into the right atrium. From the right atrium, it enters the right ventricle. Blood leaves the right ventricle by the pulmonary arteries. Completely deoxygenated blood is only found in the pulmonary arteries. The pulmonary artery branches into two, directing blood to the left and right lungs. At each lung the artery divides and divides into ever smaller vessels, which eventually become capillaries. All this capillary branching creates a greater surface area so larger volumes of blood can efficiently replace carbon dioxide with oxygen, by bringing it to the surface of the tissue.

- 3 Figure 1 shows a simplified flow chart (not to scale) of the systemic and pulmonary circuits. Label the diagram showing the left atrium, left ventricle, right atrium, right ventricle, septum and the direction of blood flow.

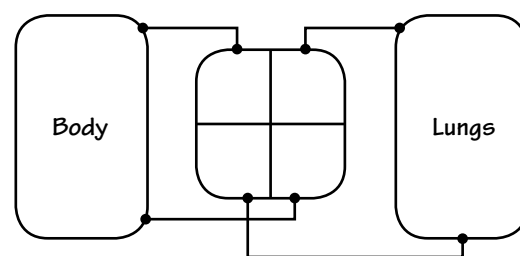


Figure 1

The stethoscope

The stethoscope is a mechanical instrument used to magnify sounds within the body, usually used for the chest region (*stethos*: chest/breast, *skopein*: to look at). Figure 2 shows a conventional stethoscope.

- 4 Outline how a stethoscope works. _____



Figure 2

As the arterial valves open and close the actions can be heard as a 'lubb-dub' sound. The 'lubb' is made when the atrioventricular valves open as blood leaves the atria to enter the ventricles. The 'dub' immediately following is the sound of the semilunar valves closing as the ventricles contract, sending the blood out of the heart.

Measuring blood pressure

Systole and diastole – The contraction and relaxation phases of cardiac muscle, commonly called heartbeat, is referred to as systole and diastole. For blood to flow through the circulatory system with sufficient pressure, the heart must send it out with a powerful push. This push is followed by a brief resting phase during which more blood enters the chambers. The arterial system is always under pressure, but the pressure increases during systole. Systole occurs when the heart muscle contracts. Diastole is the resting, refilling phase. The constant force of sending blood through the body requires the walls of blood vessels to be flexible enough to expand and contract with each thrust.

Using the large arteries, blood pressure can be measured. An average, healthy, blood pressure reading is 120/80. The first number gives the systolic (contraction) pressure. The second number gives the diastolic (resting) pressure. It is particularly important that the diastolic pressure does not rise much above 80. If so, this indicates that the heart is not getting sufficient rest between contractions.



- 5 If someone had a blood pressure reading of 125/85 would they be in danger? Explain your answer.

6 What would a blood pressure reading of 135/100 indicate?

A sphygmomanometer is used to measure blood pressure. A hollow cuff is wrapped around the upper left arm and a hand pump is used to fill the cuff with air. A stethoscope is placed below the cuff and, as the cuff tightens, the stethoscope is used to listen to the artery. There is liquid metal mercury contained in a thin glass column on the sphygmomanometer.

When the blood flow to the artery is stopped, (by the tightness of the cuff) nothing can be heard through the stethoscope. As the air is slowly released from the cuff, blood flow slowly resumes. The stethoscope begins to detect a faint clicking sound as the blood begins to flow. As soon as the sound is detected, a reading is taken of where the mercury is in the glass column. This measures the systolic pressure. As the cuff loosens further, and the pressure it applies is reduced, the clicking sound disappears. At this point diastolic pressure is measured by taking a second mercury reading. However, electronic versions of the traditional sphygmomanometer are now commonly used.

Blood pressure is measured using the arteries because they are the largest blood vessels (under the greatest pressure) and therefore give good readings. Also, pressure on the blood vascular system reduces as vessels decrease in size, making it more difficult to obtain a reading.

- 7 Mark the following statements as true or false. Then correct the false statements so that they are true.
- | | |
|--|---|
| a Arteries are under greater pressure than veins and capillaries._____ | e The device used to measure blood pressure is called a sphygmomanometer._____ |
| b The only arteries to carry completely deoxygenated blood are the pulmonary arteries._____ | f It is important that diastolic pressure be above 80._____ |
| c Systole refers to the heart's resting phase._____ | g The systemic circuit carries oxygen-rich blood._____ |
| d The pulmonary circuit is the circuit that carries oxygen-poor blood to the lungs._____ | h Carbon dioxide/oxygen exchange occurs at the capillaries of the lungs._____ |

For Questions 8–12 you must do some research.

- 8 *High blood pressure* is often a sign of potential illness. Outline what is meant by this term._____
- _____
- 9 Name some medical conditions with which high blood pressure is associated._____
- _____
- 10 Outline what is meant by *low blood pressure*._____
- 11 Name some medical conditions with which low blood pressure is associated._____
- _____
- 12 Which is more dangerous, high or low blood pressure?_____

Heart murmurs often indicate heart defects. Up to 95% of young children produce the sounds of a heart murmur at some time, but this doesn't indicate a heart defect. In most cases, the murmur disappears when the myocardium thickens and strengthens as the child grows. In cases where the murmur results from an abnormality, leaky valves, disease or other defects must be investigated. Murmurs are usually associated with leakages of blood commonly caused by either damaged or improperly closing valves. The sounds are often characterised as either 'f-f-t-f-f-t' or 'f-sh-f-sh' in association with the normal 'lubb-dub.'

- 13 Millions of people die annually from heart attacks (coronary). Briefly describe what a heart attack is. Explain the difference between the terms *coronary* and *angina*.



Adult humans have around 5 litres of blood. Human blood is made up of plasma (over 90% water) and blood cells. Plasma is a watery fluid (serum). The blood cells that are transported in the plasma carry nutrients to body cells, waste matter from body cells, antibodies, enzymes and hormones. Red blood cells make up 99.9% of blood cells – platelets and white blood cells make up the remainder.

Most oxygen is transported through the body bound to haemoglobin in red blood cells. Haemoglobin is a type of protein. Only about 2% of oxygen is transported in plasma. When blood clots, the watery plasma separates from the cells and the resulting cellular substance forms the clot.

1 Identify the following blood components from their descriptions.

- a** Pale liquid that is the major constituent of blood_____
- b** Defence protein in blood made by lymphocytes_____
- c** Disc-shaped cells, also called erythrocytes_____
- d** Gas which exists as molecules with the formula O_2 _____
- e** Gas that is a product of respiration_____
- f** Red pigment in blood cells_____

2 Read the following passage and write definitions for the words in *italics* on the lines below.

When blood contains the maximum amount of gas that it can, it is *saturated*. Normal human blood can transport about 0.2 mL of oxygen per mL of blood. Haemoglobin that has oxygen attached (oxyhaemoglobin) is a different colour to haemoglobin that has no oxygen attached (deoxyhaemoglobin). When light of specific wavelengths is passed through a blood sample, the percentage of oxyhaemoglobin can be measured. This is used to calculate oxygen saturation.

When viewed through the skin, blood that is low in oxygen appears blue. But the blood itself is not blue, even when low in oxygen.

Another method of *analysing* blood gases is with *arterial* blood gas (ABG) analysis machines. These can measure the amount of oxygen and carbon dioxide in a sample of blood by monitoring the rate of *diffusion* of these gases through artificial *membranes*, which are *permeable* to these gases. When moving through a membrane, oxygen in the blood produces an electrical current, while carbon dioxide changes the *pH* of the solution. MRI whole-body scanning can be used in *vivo* to monitor oxygen and carbon dioxide concentration.

- a** _____
- b** _____
- c** _____
- d** _____
- e** _____
- f** _____
- g** _____
- h** _____

3 Mark the following statements as true or false. Then correct the false statements so that they are true.

- a** Blood high in oxygen appears bright red, while blood low in oxygen appears darker._____
- b** The amount of oxygen in blood will be the same all around the body._____
- c** 1 mL of human blood usually contains about 0.2 mL of carbon dioxide._____
- d** The greater the carbon dioxide in a solution, the higher the pH of the solution._____
- e** In itself, plasma is an acellular substance._____
- f** Red blood cells represent most of your body's blood cells._____
- g** Water forms an important part of plasma._____



The liver is the largest of the major internal organs. It is located (roughly) under the bottom of the right side of your ribcage. The average human liver weighs just over 1.5 kg and is a dark brownish-red colour. It consists of a large right lobe and a smaller left lobe. The liver is involved in digestion, metabolism and detoxifying the body (removal of toxins).

- 1 Define *toxin*._____

The liver produces a substance called bile. Bile is used to break down fats (and fat-soluble vitamins) by emulsifying the foods we eat. Bile is a brownish-green, bitter substance. It is stored in the gallbladder.

- 2 Label the liver and the gall bladder in figure 1.
- 3 Bile emulsifies fats from the foods we eat. What does *emulsify* mean?

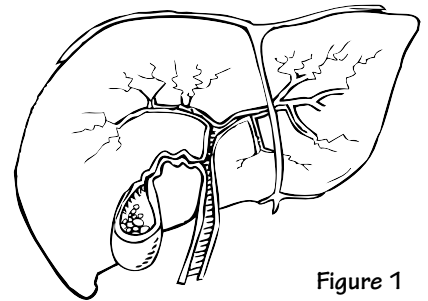


Figure 1

- 4 **Research** – Sometimes ‘gallstones’ can develop in the gallbladder. Find out what they are and what causes them to develop.

A good way to picture the liver is by imagining strings of cotton reels suspended in fluid. Each cotton reel represents a cell and there are many of these cells suspended in a pool of blood.

- 5 If you squeeze a piece of raw liver between your fingertips you will see how easily it collapses. What does this tell you about the cells?

There is a lot of activity going on among the ‘strings’ of liver cells. They work with the immune system and also play an important role in managing how our blood clots. When the liver is damaged, these strings of cells harden and become useless scar tissue. Fats and fat-soluble substances accumulate in our bodies, as do toxins, bacteria and germs. Without our livers, we would soon drown in our own poison. Liver failure kills.

- 6 The liver is involved in detoxification and metabolic activities. What do the terms *detoxification* and *metabolic activities* mean?

Paracetamol poisoning – A real danger!

It has been estimated that Australian hospitals see an average of 45 paracetamol overdoses per month. This astonishing number is not unusual in western countries. Health workers are concerned with these figures, especially since a large majority of victims – about two-thirds – are teenagers. Paracetamol can be bought without prescription, but this does not mean that it is a totally safe drug. Many people believe that if two tablets will help with a headache, four will do the job faster and better. This is extremely dangerous since the liver can only process so much of the drug at a time. This is why recommended doses are vital and are given on all medications.

- 7 Paracetamol is an analgesic. Define *analgesic*._____
- 8 Why are recommended doses given on packets of paracetamol (and other medications)?

Some victims of paracetamol overdoses took the tablets as an attention-seeking ‘suicide threat’. These people believed that the poisonous threat of the tablets would frighten their loved ones without doing themselves any real harm. Unfortunately those who survive learn just how deadly paracetamol can be. Other users ‘pop them like lollies’ for the smallest ailment, again not realising that paracetamol can kill. Overdose can sometimes be treated with an antidote (N-acetylcysteine) but it must be given within a very narrow window of time, or liver failure occurs. When the liver fails the patient either dies or an emergency transplant is needed if a matching liver is even available. However, the antidote has proven dangerous to asthmatics and cannot help if it is not given quickly enough.



- 9 Paracetamol is an *antipyretic*. Antipyretics are drugs that reduce fever by overriding the brain's instructions to the body to heat up. Would an antipyretic lower the body temperature of a person who did not have a fever? Explain.
- 10 **Research** – Paracetamol is sometimes a preferred alternative to aspirin. Outline possible reasons for this.
- 11 Discuss this statement: *Prevention is always better than cure.*

Alcohol and the liver

One of the most common alcohol-related diseases is cirrhosis (pronounced *sir-oh-sis*).

Cirrhosis causes the liver to enlarge, then shrink. After shrinking, it hardens and the once blood-rich liver tissue is replaced by hard, dead, scar tissue. The replacement of live tissue with scar tissue results in the liver being smaller (overall), and therefore less effective. One sign of poor liver function is a condition called jaundice, or 'yellow' jaundice because it causes the eyes and skin to turn yellowish. Continual alcohol consumption by someone with cirrhosis leads to death by liver failure. Jaundice occurs when bile from the gallbladder makes its way into the bloodstream. Hepatitis C, another major cause of liver failure, also causes jaundice.

- 12 Figure 2 shows the structure and formula of ethanol (alcohol). Name the elements present.

Many people show no symptoms of cirrhosis until the damage has been done.

Symptoms include fatigue, exhaustion, a bloated 'barrel' belly, difficulty sleeping and forgetfulness (often due to the build-up of toxins in the brain), nausea, weight loss, abdominal pains, the appearance of spider angiomas under the skin (blood vessels that look like spiders), excessive bruising and bleeding (due to the blood-clotting mechanism being affected), lost appetite, general weakness and lethargy.

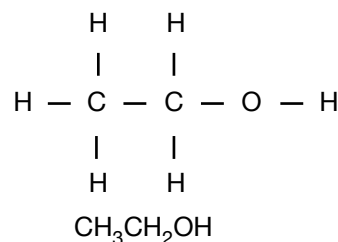


Figure 2

- 13 **Research** – Find out some other complications caused by cirrhosis of the liver.

- 14 There is no vaccine for Hepatitis C (Hep C), one of the major causes of liver disease. Find out two ways in which a person can contract the virus.

- 15 Mark the following statements as true or false. Then correct the false statements so that they are true.

- | | |
|--|---|
| a When symptoms of cirrhosis appear, the liver has already been permanently damaged._____ | i Ignorance of the dangers of paracetamol lead to many overdoses._____ |
| b The liver contains a large left lobe and a small right lobe._____ | j Hepatitis C can be prevented with a vaccination._____ |
| c Teenagers represent a large proportion of paracetamol overdose victims._____ | k Paracetamol has a very different chemical structure to aspirin._____ |
| d Fats are stored in the gallbladder._____ | l Alcoholism causes the liver to swell and then shrink._____ |
| e Antipyretic drugs override the brain's instruction to raise body temperature._____ | m Jaundice is caused by the presence of bile in the bloodstream._____ |
| f Liver failure kills many people every year._____ | n The liver is the second largest of the body's major organs._____ |
| g Bile is a substance that breaks down proteins and protein-soluble substances._____ | |
| h Hepatitis C is a major cause of liver failure._____ | |



The kidneys are the two main organs of excretion. Excretion is the process of removing waste from the body. Your kidneys are paired and each is about the size of your fist. They are found in the lower back on each side of the spine. The left kidney is located just under the liver and the right kidney is not far from the spleen. Because the liver is large and needs more space, the left kidney is a little bit lower than the right. Kidneys filter blood, removing unwanted substances. As part of homeostasis they regulate the level and contents of body fluids.

Blood filled with impurities is carried to the kidney by the renal artery. The renal artery branches into smaller arterioles and then into capillaries which will later join together to empty purified blood back into the renal vein. The outer part of an organ is called the *cortex* and the inner part is called the *medulla*. Nephrons are the filtering units of the kidney. They are housed in hollow cavities in the medulla. Nephrons are tiny units and there are approximately one million in each kidney. Each nephron is made up of a renal corpuscle and a kidney (uriniferous) tubule.

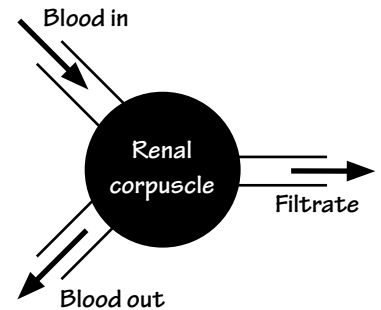


Figure 1

- 1 Blood enters a kidney through a renal _____.
- 2 Blood leaves a kidney through a renal _____.
- 3 What is a *tubule*? _____

Each renal corpuscle is made up of a glomerulus (a ball of capillaries) and a Bowman's capsule (a thin-walled sac surrounding the glomerulus). Blood is made of cells and plasma. Plasma makes up about 55% of blood's composition. It is a straw-coloured substance made mostly of water, containing inorganic salts, proteins and other substances.

- 4 Suggest why plasma is made almost entirely of water.

Blood in the glomerulus is under much greater pressure than blood elsewhere in the body. The higher pressure causes the blood flow to slow down and 'squeeze out' much of the blood's plasma. This slowing of blood enables the filtration system to remove impurities much more effectively. The plasma is squeezed into Bowman's capsule, where filtration takes place. Once impurities are removed, most of the plasma (about 99%) is reabsorbed by the nephrons to be recirculated throughout the body. The remainder is used to carry waste substances out of the body.

- 5 A cavity enclosed by a membrane within a living organism – what does this define? _____

The kidney tubule is made up of the proximal convoluted tubule, the loop of Henlé and the distal convoluted tubule. The glomerular filtrate (water, amino acids, minerals, glucose and urea) is carried out into the proximal convoluted tubule.

- 6 Define *proximal*. _____
- 7 Define *convoluted*. _____

Most of the useful dissolved substances are reabsorbed into the bloodstream as the filtrate passes through the loop of Henlé and the distal convoluted tubule. The tubules join to form a urine collecting duct and these collect into the ureter.

- 8 Name three substances in the filtrate that might be desirable to reabsorb into the blood stream.

- 9 What is meant by *distal*? _____
- 10 What do we call the glomerular filtrate once it reaches the distal tubule? _____
- 11 On the section of kidney on the next page (figure 2), label these parts:
 - a cortex
 - b medulla
 - c pelvis
 - d ureter



12 On the section of kidney (figure 2), draw in the:

a renal artery.

b renal vein.

13 Briefly summarise the function of the kidneys.

14 Mark the following statements as true or false. Then correct the false statements so that they are true.

a The cortex is the inner part of an organ. _____

b Homeostasis refers to maintaining a constant environment in the body. _____

c Plasma is almost entirely made of water. _____

d Blood filled with impurities is carried to the renal vein. _____

e About 1% of the water entering the kidneys is reabsorbed after filtration. _____

f The loop of Henlé joins the ureter. _____

g Blood in the glomerulus is under high pressure. _____

h Bowman's capsule is a thin-walled sac surrounding the glomerulus. _____

i There is a pair of kidneys found on each side of the spine at the lower back. _____

j The nephron is the functional processing unit of the kidney. _____

15 Research – On top of each kidney sits an *adrenal gland*. Find out what the adrenal gland is. _____

16 Research – Find out what a diuretic is, and how it affects the kidneys. Give an example of a diuretic.

17 It is important that urea and other compounds found in urine do not build up in our bodies. Why?

18 What is a *filtrate*? Give an example of an everyday filtrate. _____

19 Where is liquid waste stored in the body until it can be expelled? _____

20 Research – Find out if it is possible to survive with only one kidney, and explain your answer.

21 *The extraction, removal and release of unwanted substances from the body (the products of metabolism)* – what process does this define?

22 Research – Find out one cause of kidney disease/failure. _____

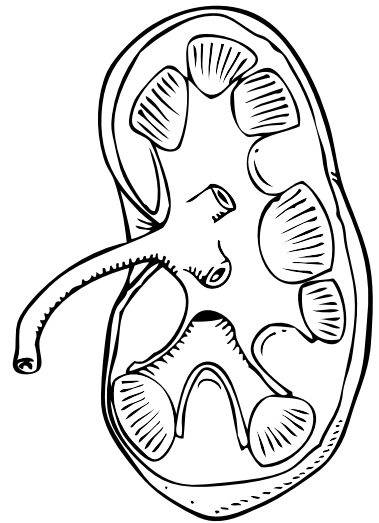


Figure 2



The nervous system uses electrical impulses to control the body. The nervous system is necessary for instant communication and reaction. The communications are short, fast and tend to be limited to a small area. It is an important control system but not the only one.

There is another important control structure – the endocrine system. This is a slower system, with longer term effects. The endocrine system is a network of glands (and cells) in animals that send hormones (Greek *hormaein*, meaning ‘to set in motion, to move’) throughout the body. Plants do not have an endocrine system.

Hormones are ‘chemical messengers’ that cause or prevent specific reactions. Areas in which hormones exert their effects are reproduction, growth and development, control of the internal environment and regulation of energy production. Some hormones, such as those produced in the thyroid gland (and growth hormones), affect all the cells of the body, not just limited targets.

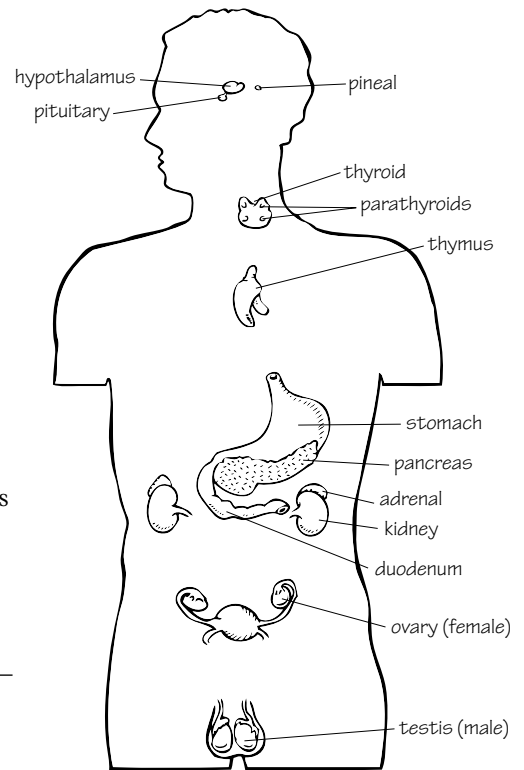


Figure 1

- 1 What do the nervous system and the endocrine system have in common?

- 2 Name two differences between the nervous system and the endocrine system.

- 3 Name three areas over which hormones exert an influence. _____

Hormones are secreted (released) by glands into the bloodstream which then carries them to the target sites. Figure 1 shows the location of the major glands (and some other organs). Hormones control and regulate the body’s activities by sending messages to tissues, cells or organs. Only small amounts of hormones are required to perform these functions. Hormones have specific protein receptors or sites to which they bond. The receptors can ‘recognise’ what they must bond to as they travel through the bloodstream. When bonding occurs, a reaction is triggered.

The pituitary gland is sometimes called the ‘master gland’ because the hormones it secretes affect so many other glands. Figure 2 shows the location of some parts of the brain and associated glands.

Most glands secrete substances through ducts. The endocrine system does not use ducts, but secretes the hormones directly into the bloodstream. The effects of one hormone on a gland or organ can lead to effects on another. When the system is working properly, the organs work well. When an imbalance occurs, an overproduction or underproduction of hormones can occur. Imbalances can be fatal, so they must be treated by a doctor.

You will encounter the terms *hyper* and *hypo* in your studies. The term *hyper* means ‘overactive’ or ‘excessive’. *Hypo* is the opposite, meaning ‘under active’ or ‘insufficient’.

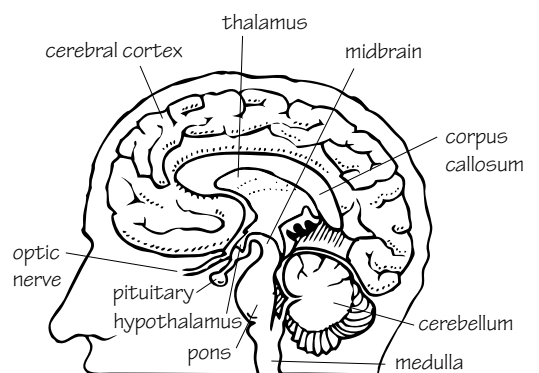


Figure 2

- 4 Cross out the incorrect terms in *italics*.
 - a A hyperactive endocrine gland produces *too much* / *too little* hormone for the body’s needs.
 - b Hypoactive glands are producing *too much* / *too little* of the required hormones.
- 5 What is the name of the system that controls the body through the release of hormones? _____
- 6 What is the general name for organs which secrete hormones? _____
- 7 Which part of the brain controls a large proportion of hormone production? _____



Hormone secretion – The output of a hormone is usually regulated – the effects they produce often slow down or stop output of the hormone once sufficient amounts are present. Many hormones are released in response to stress. Others show a daily, or circadian, pattern (*circadian* from Latin meaning ‘around a day’), often peaking in the early morning – probably to prepare the body for the stress of getting up. The symptoms of ‘jet lag’ may be due to the body getting out of step with hormonal rhythms.

Adrenal glands – There are two adrenal glands – one located above each kidney (*ad* meaning ‘near’, *renal* meaning ‘kidney’). They secrete the hormone epinephrine (adrenaline) at times of excitement or danger. Adrenalin (see figure 3) makes the heart beat faster, increases the breathing rate and also diverts blood supply to muscles. When the stimulus is over, levels of the hormone drop back to normal. Adrenalin is often called the ‘fight or flight’ hormone.

8 Name the elements present in adrenaline.

9 How do hormones reach every part of the body?

10 List three effects of adrenalin.

11 How many adrenal glands are there?

12 Where are the adrenal glands located?

13 What may cause ‘jet-lag’?

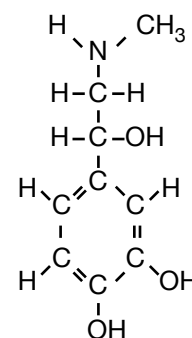


Figure 3

The **pituitary** gland is located at the base of the brain. Despite its small size it performs many crucial functions. It releases hormones which control growth and reproduction, and manufactures other hormones which, in turn, control many other glands. When people grow too big (giantism) or too small (dwarfism), a defective pituitary is often responsible. The pituitary releases chemicals that regulate lipids and carbohydrates. Thyroid-stimulating hormone (TSH) is released by the pituitary, and is a vital part of the endocrine’s feedback system. TSH levels are part of the system that is responsible for triggering much of the thyroid’s activities. When something is in short supply or in excess, the pituitary recognises this and adjusts its output accordingly.

The **thyroid** gland is the largest of the endocrine glands. It is located in the neck below the larynx. It secretes a hormone which is essential for the maturing of the central nervous system. An overactive thyroid gland is called hyperthyroidism (Graves’ disease, thyrotoxicosis). Common symptoms include heart palpitations, fatigue, irritability, hand tremors, weight loss, fainting and impaired vision. If left untreated, this condition can be fatal. An underactive thyroid, hypothyroidism (hyxedema, cretinism), is less dangerous to adults though should be treated. (Many people have hypothyroidism without realising it.) Symptoms include weight gain, irritability, cramps, depression, constipation, dry hair and hair loss. Hypothyroidism in babies must be treated if they are to develop properly.

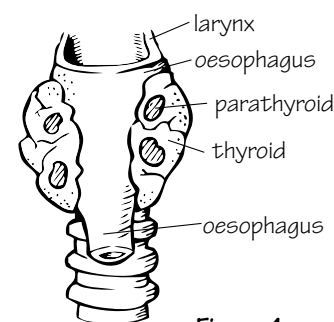


Figure 4

14 How is the larynx involved in speech?

15 The oesophagus is also known as the _____.

16 A collective term for all the complex, closely-coordinated chemical reactions occurring inside an organism – what does this define?

17 What does TSH stand for? Which gland releases TSH?

18 Why is the pituitary gland called the ‘master endocrine gland’?

19 Name two specific roles played by the thyroid gland.



When medications fail, hyperthyroidism is often treated using radioactive iodine. The patient swallows a radioactive iodine tablet and the iodine is taken up naturally by the thyroid. The radiation destroys part of the thyroid. Within a few days, the larynx area is no longer radioactive.

- 20 Would the thyroid still be overactive after radioactive iodine treatment? _____
- 21 How could destroying part of the thyroid be useful? _____
- 22 What would be the main side effect of this treatment? _____

The parathyroid glands manage calcium levels in the blood.

Pancreas – This is found near the stomach and is a large gland that secretes pancreatic juice, which is used in digestion. It also produces the hormones insulin and glucagon. A constant blood sugar level is important in the body. When it drops too far, the pancreas produces glucagon to raise it again. If a high glucose level arises, insulin is produced to lower the level. People who suffer from the disease Type I Diabetes (Diabetes Mellitus) have a deficiency of insulin. The disease cannot be cured, but it can be controlled by dietary restrictions and/or injections of insulin.

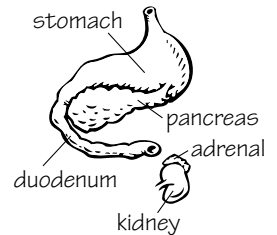


Figure 5

Testes and Ovaries – Reproductive function in both sexes is controlled by hormones from the pituitary gland and the steroid hormones from an ovary or a testis. In women, fluctuations of these hormones govern each menstrual cycle, stimulating ovulation and preparing the uterus (womb) for pregnancy. The sex steroids also bring about the development of secondary sexual characteristics – beard growth, deep voice and muscle increase in males, breast development and rounding of hips in females.

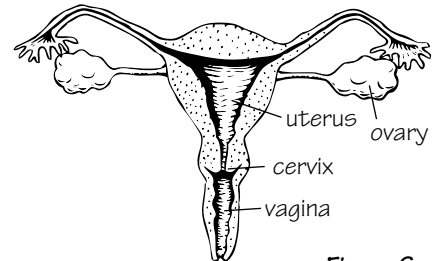


Figure 6

- 23 Name the gland that produces insulin. _____
- 24 How is type 1 diabetes managed? _____
- 25 What is another name for the *uterus*? _____
- 26 Outline the effects of male and female sex steroids. _____
- 27 Mark the following statements as true or false. Then correct the false statements so that they are true.
 - a The two main control systems in the body are the nervous and the endocrine systems. _____
 - b The endocrine system operates using a series of ducts. _____
 - c Hormones are molecular chemical 'messengers'. _____
 - d The thyroid is the largest endocrine gland. _____
 - e Of the body's two main control systems, the endocrine has the fastest reaction time. _____
 - f The substance epinephrine is often called adrenaline. _____
 - g The prefix *hypo* means 'under' (active.) _____
 - h Graves' Disease is the result of an overactive thyroid. _____
 - i The pituitary gland is located above the kidneys. _____
 - j The kidneys are responsible for the 'fight or flight' response. _____
 - k Hormones travel through the bloodstream then attach to target receptors. _____
 - l The thyroid produces thyroid stimulating hormone (TSH). _____
 - m A defective pituitary gland can lead to too little or too much growth. _____
 - n The nervous system can be useful for getting the body out of instant danger. _____
 - o Only small amounts of hormone are required for a reaction. _____
 - p The endocrine system coordinates the body's metabolic activities. _____
 - q Plants have a slightly different endocrine system to animals. _____



Cut out the boxes and paste them under the correct headings (shown below) in your workbook. Line up each endocrine gland with its location (where it is found) and major function (what it does). You may need to use a text book to help you. (Note the major hormonal functions are not complete lists.)

Name	Location	Function
Thyroid		Rate of metabolism
		Pelvic area
Produces insulin, regulates sugar levels		Testes
		Produces adrenaline
	Base of brain	
Pancreas		Ovaries
		Close to stomach
	Neck, near larynx	
Above the kidneys		
		Regulates insulin production, blood pressure, skeletal growth
Produces oestrogen, initiates secondary female sexual characteristics		
	Lower pelvic area	
Adrenal gland		Pituitary
	Produces androgen, initiates secondary male sexual characteristics	



The nervous system uses electrical impulses to control the body. There is another important control system – the endocrine system. This system works (more slowly than the nervous system) by releasing chemicals called hormones. Hormones are special chemical ‘messengers’ which control various activities inside an organism. Human hormones are secreted by endocrine glands in response to their environment. Some hormones only act on specific body parts; others cause a more general response. Areas in which hormones exert their effects are reproduction, growth and development, control of the internal environment and regulation of energy production.

Hormone: a product of living cells that circulates in body fluids and produces a specific effect on the activity of cells remote from its point of origin

- 1 What transports hormones from their point of manufacture to where they are used? _____

The output of a hormone is usually regulated – the effects they produce often slow down or stop output of the hormone.

- 2 Write the name of a process by which output of hormones can be regulated. _____

Many hormones are released in response to stress, while others show a daily (circadian) pattern, often peaking in the early morning, probably to prepare the body for the stress of getting up. Aldosterone (a corticosteroid) has the formula $C_{21}H_{28}O_5$. It helps in the regulation of the salt and water balance of the body by affecting proximal tubules.

- 3 Name the main elements present in aldosterone. _____

- 4 Which organ contains *proximal tubules*? _____

Aldosterone is secreted by the adrenal gland. There are two adrenal glands – one on each kidney.

The level of hormone changes in response to changes in blood pressure and the levels of electrolytes such as sodium (Na^+) and potassium (K^+) in plasma. If the kidneys excrete too much sodium, an excess of body fluid is also lost. Heart function can be affected if potassium levels go outside certain levels.

- 5 Electrolytes are *ions*. What is an ion and how do ions help with reactions? _____

Hypoaldosteronism is a condition where aldosterone isn’t secreted due to damage to the cortex of the kidney. It causes dehydration (excessive water loss), a drop in blood pressure and excessive loss of sodium.

- 6 *Hypo* and *hyper* are two terms often confused. Explain the difference between them. _____

- 7 If a doctor wanted to test your aldosterone levels, what sort of test would be performed and why? _____

- 8 Define *endocrine system*. _____

- 9 On the lines on the next page, write meanings for the words in *italics* in this passage.

Addison’s disease (also called *chronic* adrenal insufficiency) results from the failure to *secrete* aldosterone. *Symptoms* include a high urine output which gives a low blood volume. If blood pressure falls to low levels, this can cause heart failure. Hormone replacement therapy is needed, as without treatment sufferers will die. Pure aldosterone can be *synthesised* from other steroids but is very expensive. A synthetic replacement hormone, fludrocortisone (Florinef), is used to treat this condition but needs to be *monitored* carefully.



- a _____
- b _____
- c _____
- d _____
- e _____

- 10 List possible symptoms if too much fludrocortisone is given to a person with Addison's disease.

- 11 Fludrocortisone produces *anti-inflammatory responses*. What does this mean? _____

- 12 One of the effects of Addison's disease is *anaemia*. What is anaemia and how can anaemia be dangerous?

Feedback loops – Figure 1 shows an incomplete homeostatic feedback system/loop.

- 13 Complete the feedback mechanism diagram using these terms:
blood reaches organ, changes level of hormone production, sends message to, signal sent to brain

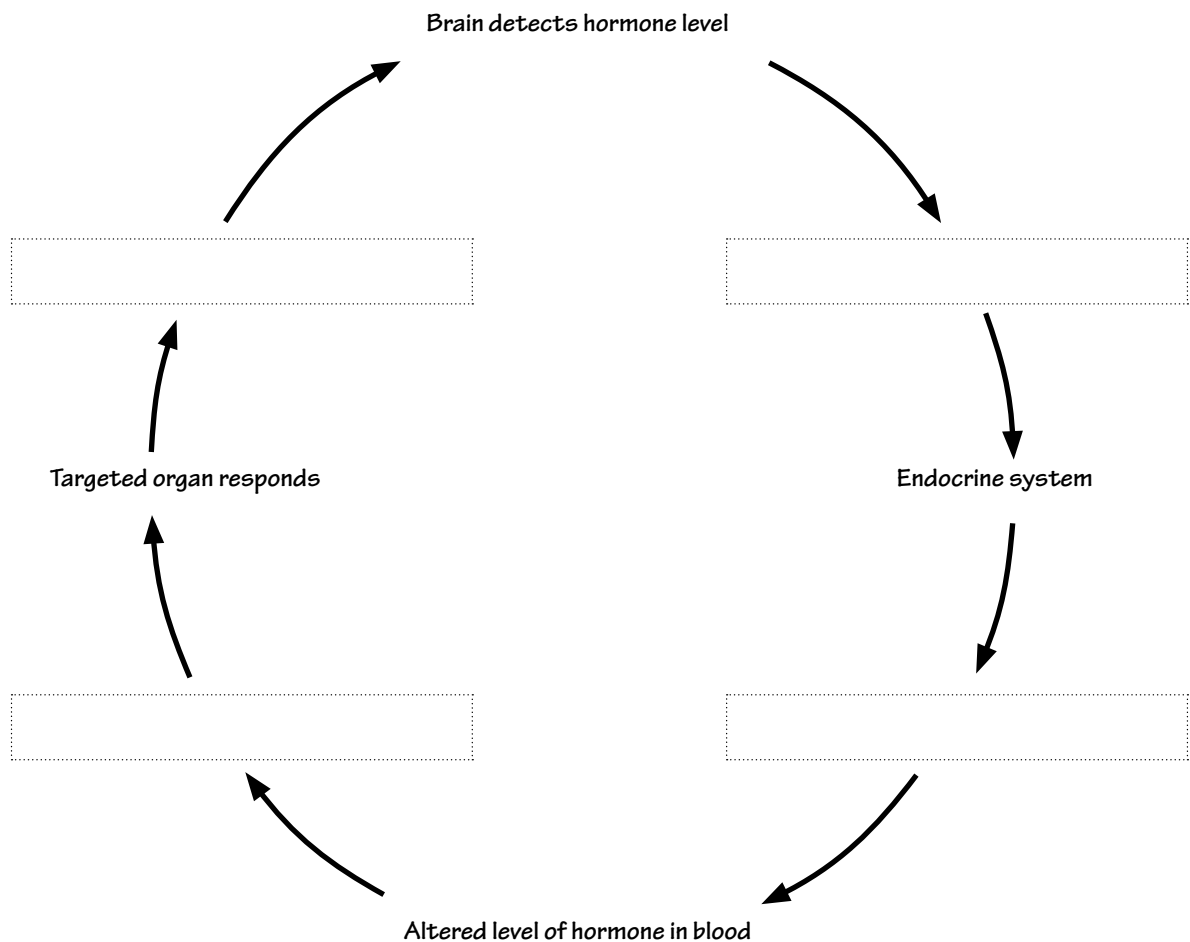


Figure 1



You may need a text book to help with this section

The nervous system connects every part of the body to the brain by nerves. The brain controls the body – it sends (and receives) signals along these nerve fibres. The electrical impulses sent along the nervous system travel very quickly, e.g. from the big toe to the brain in a few thousandths of a second. Nerves can be very short or over a metre long. The nerves along which signals travel are made up of bundles (varying from a few hundred to a few thousand) of long, thin cells called neurones.

Neurones are specialised nerve cells that transmit signals by electro-chemical changes in their membranes. They are ‘long distance’ nerves. There are three major types of neurones – sensory neurones, motor neurones and interneurones. Neurones are made up of a cell body, nucleus, axons, dendrites and protective and supportive structures. The gap between the axon of one neurone and the dendrite of the next is called a synapse. Electrical impulses ‘jump’ across each synapse.

- 1
- Draw a diagram showing the cell body, nucleus, myelin sheath, axon and dendrite of one neurone and a synapse and the dendrite of the next neurone.

- 2
- Write your own definition for *nerve* (in biology terms)._____

Dendrites are short extensions of the cell body. They receive stimuli and carry them to the cell body. Axons are much longer and there is usually only one axon per neurone. The axon carries nerve impulses away from the cell body to other cells. Axons are branched. When an impulse reaches the end of the axon branches, chemicals are released (neurotransmitters) to the dendrite of the next cell. When the neurotransmitter reaches this dendrite, a fresh electrical signal is generated and this travels through this next cell, and so on.

- 3
- Complete the table.

Part of neurone	Function
axon	a _____
b _____	insulates axon so electrical signal can travel
nucleus	c _____
d _____	carries stimuli to cell body of neurone
synapse	e _____
f _____	chemical released from end of axons



Sensory neurones (*afferent*, meaning ‘carry towards’) respond to a stimulus and send a message in the form of an electrical impulse to the central nervous system. They are found all over the body. Interneurones (connecting neurones) link the sensory and motor neurones. There are vast quantities of these in the grey matter of the Central Nervous System. Motor neurones (*efferent*, meaning ‘carry away’) take the message away from the Central Nervous System and cause a response. This response can be to cause a muscle to contract or a gland to secrete a hormone.

Central Nervous System (The brain and the spinal cord)

The brain controls our actions and feelings; it consists of billions of neurones, interconnecting with each other. It is protected by the skull. The brain is connected to the spinal cord, which is protected by the backbone.

The brain is a highly developed, dense mass of nerve cells that forms the upper end of the central nervous system.

An adult brain weighs 1400 g on average. The three parts of the brain that we will look at further are the cerebellum, cerebrum and medulla oblongata.

The cerebellum is located between the brain stem and the back of the cerebrum, and in humans is formed of two lateral (side) lobes and a median lobe, and it is round and wrinkly. It is chiefly concerned with managing the motor functions and unconscious control of muscles, e.g. co-ordination of movement and body balance.

The cerebrum is the enlarged upper part of the brain and is made up of two hemispheres and connecting structures. The cerebrum is the most recently evolved part of the brain and is the part we usually visualise when thinking of the brain. It makes up over three-quarters of the brain’s weight. The cerebrum is highly folded. It processes information from the eyes and ears and is responsible for intelligence, intellectual and creative skills and memory. Apart from the higher thought processes, it also controls conscious movements.

The medulla oblongata is an extension of the spinal cord and controls vital involuntary functions.

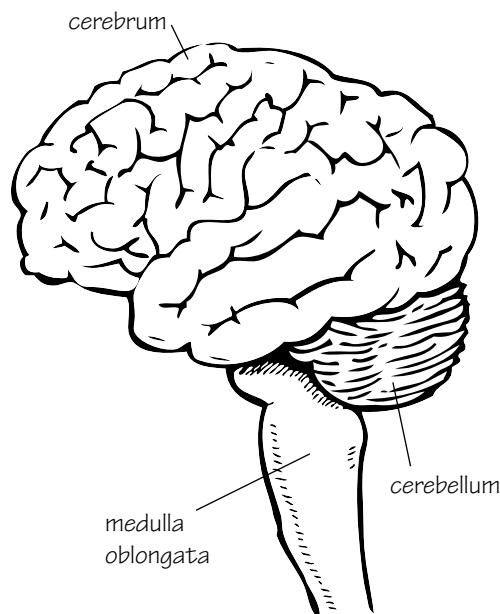


Figure 1

- 4 Name the major components of the Central Nervous System. _____
- 5 What is a *hemisphere*? _____
- 6 What is another name for a *neurone*? _____
- 7 What is a *lobe*? _____
- 8 What are the major constituents of the brain?

- 9 Suggest what the advantage of a highly folded cerebrum might be. _____

- 10 Briefly explain how the brain and spinal cord work together. _____

- 11 Suggest why people who are stressed use expressions such as ‘my nerves are shot’ and ‘I was a nervous wreck’.



The average adult brain weighs about 1.4 kg and fills most of the space in the skull (cranial cavity). Of all the body's organs, the brain is the most complex. We can explain how the heart, liver, kidneys and other major organs function, but there is still much about the brain and how it works that we don't understand.

- 1 Brainstorming** – We know less about how the brain works than any other organ. Suggest possible reasons for this.

We rely on the brain to quickly process and/or recall information. If you walk into your house at dinner time and smell fried onions and meat, you know, without seeing it, what you will have for dinner.

- 2** List three other things that we take for granted, thanks to the brain. _____

Alcohol and the developing brain

Alcohol and cigarettes are known for their damaging effects on the health of their users and on the health of developing embryos. We also know that abuse of these substances leads to the destruction of brain tissue (and the tissues of other major organs). However, evidence now suggests that these substances are damaging to the brains of adolescents and teenagers in a way previously unrealised. Until the age of eighteen years, the brain is still undergoing various stages of development with much of the 'wiring', in the form of neurones, yet to be completed – or indeed, many of the neurons are yet to be created. For this reason, the brain of an individual this age has often been referred to as 'a work in progress'.

- 3** It is illegal to sell alcohol or cigarettes to individuals under the age of eighteen. Why? _____

What is a neuron?

If you consider the brain as the supercomputer of the Central Nervous System (CNS), then neurons are its wiring. Most adult (human) brains contain about 10 billion neurons (*neuro*, meaning 'to do with nerves and the nervous system').

Neurons are specialised cells that differ from other cells in that they also have 'branches' protruding from them. The main part of the cell is the cell body. The branches are called dendrites and axons. Dendrites and axons form part of the interconnecting network through which the Central Nervous System transmits (sends) and receives information. Information is transmitted via the neurons through the CNS. The 'information highways' formed by neurons transmit their messages using electrical impulses. The impulses always travel in one direction.

The dendrites, which are short branches, generate the impulses and carry them towards the cell body. The impulses pass through the cell body and out again through the axons. Therefore, axons carry nerve impulses away from the cell body. Unlike dendrites, the axons are much longer and there is usually only one axon per cell body. Axons branch at their ends. The electrical impulses occur at incredible speeds and between many neurons at once.

- 4** Use the information above to label the neuron shown in figure 1.

Of the billions of neurons that the adult brain is intended to have, many are still being formed and connected during the adolescent and teen years. The effects of alcohol on this work can be devastating.

Alcohol prevents the formation of neurons, leaving the individual without them for life, while also 'short circuiting' many of those which have been created and are still being connected. Alcohol consumption has also been linked to increased depression and suicidal tendencies among adolescent and teenage drinkers.

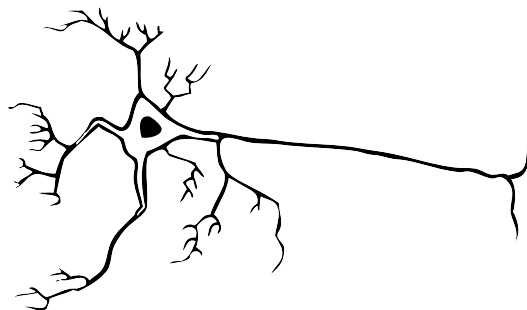


Figure 1



- 5 Use these terms to complete the passage below: *end, one, long, dendrites, impulses, many, axon*.
Electrical _____ travel in _____ direction. They arrive at the cell body via the _____ and leave it via the _____. There are _____ branches of dendrites extending from the cell body, whereas there is usually only one _____ axon. The axon is branched at its _____.
- 6 Use the information in this worksheet to complete the concept layer map (figure 2).

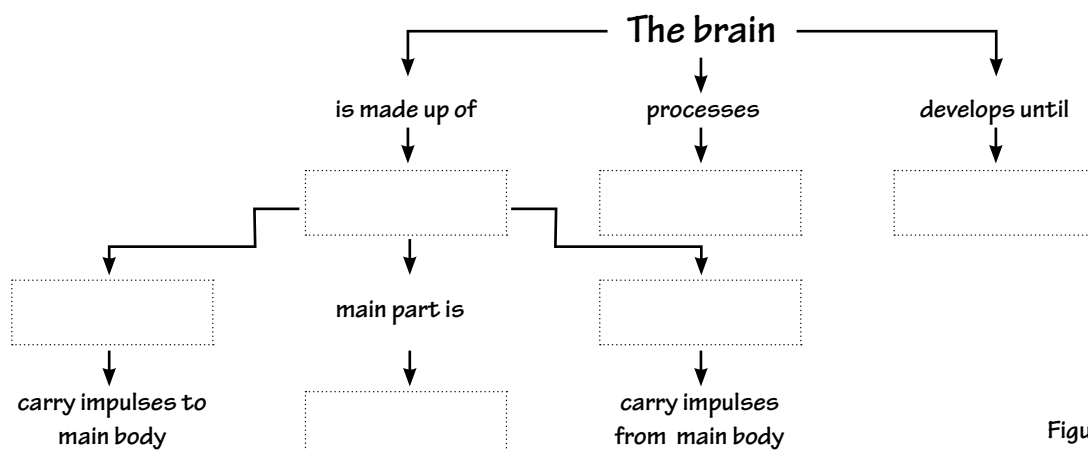


Figure 2

- 7 Alcohol and nicotine are neurotoxins. If we consider the components *neuro* and *toxin* separately, what would a neurotoxin be defined as?

General structure of the brain

The brain is much more complex than this unit will describe, but we will begin our study with a brief mention of the basics. Although the brain works as an interconnecting unit, it can be divided into three sections: the forebrain (front or anterior area), midbrain (middle area) and hind brain (back or posterior area). These divisions indicate areas of the brain, but remember that despite these divisions, the brain works as a unit.

A fissure is a crack or split and the Fissure of Rolando is a region that divides the forebrain (frontal lobe) from the posterior area. The fissure was named in honour of Italian anatomist Luigi Rolando (1773–1831). There are three major regions in the human brain: the cerebrum, the cerebellum and the brainstem. The cerebrum is the part of the brain that comes to mind when we imagine a brain. It is the largest part, divided into two hemispheres (left and right) and is deeply folded. Our brains are often referred to as ‘grey matter’ because this large area is made of grey tissue. The hemispheres are the main centre of the brain and their folding increases the cerebrum’s surface area. If we were to spread the outer layer of the cerebrum out flat, it would cover a page of newspaper. The cerebrum receives sensory information and is responsible for ‘higher’ mental activity (e.g. ‘deep’ thought, calculations) and voluntary activity (those we choose to do as opposed to involuntary, such as heartbeat, breathing, etc.).

- 8 What does *sensory* mean? Give an example of sensory information we detect. _____

- 9 **Research – Eponymous** means ‘of a person’. Many scientists, such as Luigi Rolando, have been honoured by having parts of the body or a body process named after them. Find two other examples of this.

The frontal lobe is the ‘front’, or anterior, area of the brain (directly behind your forehead). Frontal lobes play important roles in reason, speech and emotions. They are responsible for decision-making, planning, goals and aspirations, imagination and dreaming, self-control, impulses and other behaviour. They are also important in long-term memory development, recollection and the ability to learn. The frontal lobes change during adolescent and teen years and alcohol can permanently affect their development. Our ability to determine whether or not a given behaviour is acceptable occurs in this area. Alcohol dulls this sense, making self-control and other behaviour less disciplined. Details that are usually organised become confused. Under the influence of alcohol, order becomes disorder, and restrained and controlled behaviour becomes self-indulgent and excessive.



The female reproductive system has several functions. It is designed to produce female gametes, called oocytes (more commonly called eggs) and to produce important hormones that are required for the normal growth and development of the female. Unlike the male system, the female reproductive system is also designed to house and nourish the fertilised egg as it develops into an embryo until the woman gives birth.

Eggs originate in the paired ovaries. The ovaries are round organs which are essential to reproduction (Latin *ovum*, meaning ‘egg’). Approximately every month, the ovaries produce and release an egg (sometimes more than one). The ovaries are also responsible for producing the important sex hormones oestrogen (also spelt ‘estrogen’) and progesterone. Oestrogen produces the changes necessary for girls to develop into women, and triggers the onset of menstruation.

When a fertilised egg is divided into two, identical twins result. Sometimes the ovary releases two eggs at once and each is fertilised by a different sperm.

- 1 **Research** – What name is given to the twins that result from this second process? _____
- 2 Why are these twins not identical? _____

An egg is released from an ovary on a regular basis, usually every 21 to 28 days, depending on the individual. It then travels along the oviduct (Latin *ductus*, ‘to lead’; also called uterine tube or Fallopian tube) where it can be fertilised with the male sperm if it is present.

The ends of the oviducts have funnel-shaped, almost flower-like, projections (called fimbriae) which catch the egg and draw it in. Once inside, the egg travels along the oviduct, where fertilisation usually takes place. Over the period of the month, the lining of the uterus, called endometrium, gradually thickens. If fertilisation did not occur, the lining is shed and a new one begins to develop shortly after. This monthly cycle is called menstruation (Latin *menses*, meaning ‘month’) or ‘periods’. Unlike males who use the urethra to release both sperm and urine, females have a separate canal for reproductive purposes and use the urethra only for urine.

The fertilised egg will become embedded in the wall of the uterus, where it grows. When an embryo is developing inside it, the uterus is commonly called the womb. During the first eight weeks after conception the developing baby is called an embryo. At this stage all internal organs have begun to develop. From eight weeks until birth, the baby is called a foetus. At eight weeks, bones have also begun to develop.

The uterus is a hollow muscular organ that will expand as the foetus grows. When a woman is not pregnant, the uterus is usually about 7 cm long. An organ called the placenta will develop and be the ‘go-between’ for mother and baby. The placenta uses an umbilical cord to transport nutrients and waste to and from the embryo. It nourishes the developing embryo by allowing nutrients and hormones to pass through it from the mother, and removes the embryo’s waste products, also through the mother.

A thin membrane will seal the uterus when pregnancy occurs. The baby then develops in an amniotic sac of fluid, designed to cushion it, prevent dehydration and aid in its development with some nutrients. The amniotic fluid is commonly called ‘waters’. When the woman is ready to give birth, the ‘waters break’ as a result of the membrane rupturing. The baby then travels down the birth canal, the upper part of which is called the cervix (Latin *cervix*, meaning ‘neck’). The cervix is at the mouth of the uterus. It dilates to 10 cm to assist birth. The narrower cervix leads to the vagina, the lower part of the birth canal and the organ for sexual intercourse. The opening to the vagina is called the vulva, and through the vulva, the baby enters the world.

- 3 **Research** – Find out what the average weight and length of the placenta is at birth.

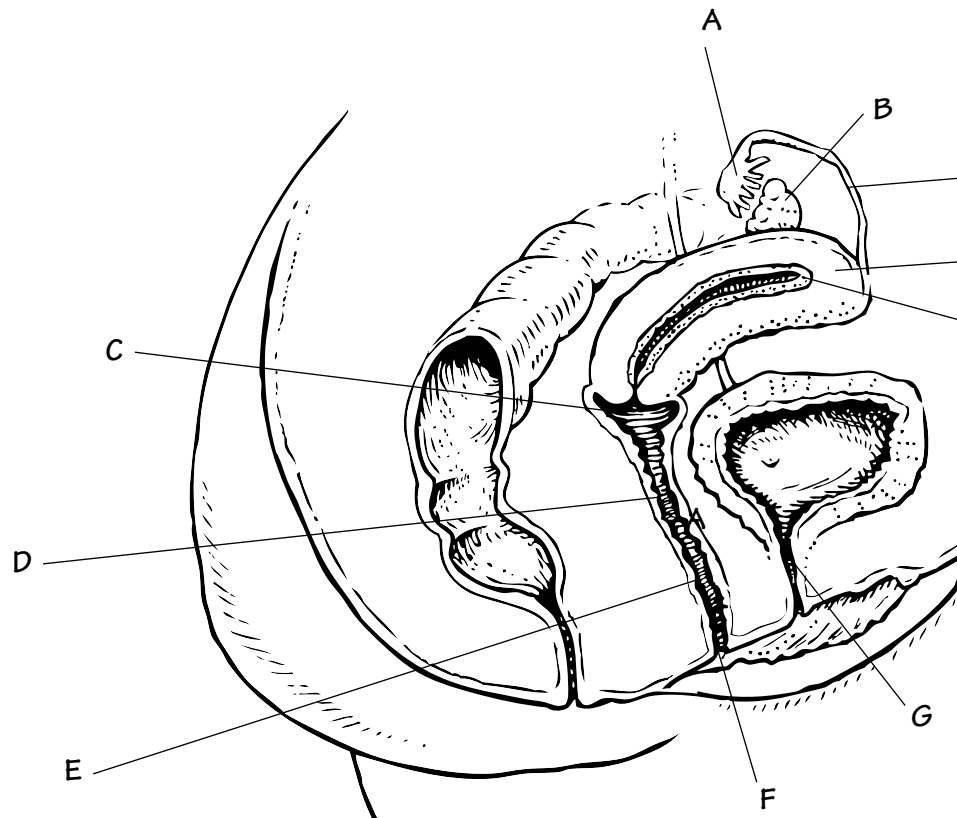
- 4 Suggest why pregnant women feel the need to urinate when their bladders aren’t full. _____

- 5 **a** Would the female reproductive system be considered more complex than the male? Why? _____

- b** When a woman is ready to give birth, she feels contractions (labour pains). Find out what causes the pains and why they are helpful to the delivery of a baby.



- 6** Label the side view illustration (figure 1) of the female reproductive system. Include a brief note describing the functions of each part.



- 7** Label the parts of the front view of the female reproductive system (figure 2).

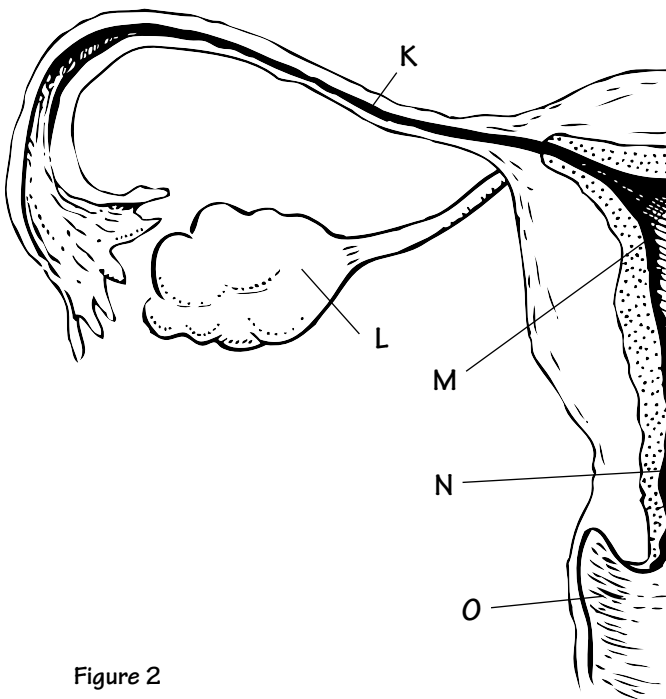


Figure 2

- 8** Reproduce figure 2 as a mirror image, on the left side of the page.
- 9** Draw a red star in the area where fertilisation usually takes place.
- 10** Draw a blue star in the area where the fertilised egg becomes embedded in the uterus.



The male reproductive system produces male gametes (sex cells) called sperm (spermatozoa), as well as important hormones that are required for normal growth and development. The phase from childhood into adulthood, in both sexes, is called puberty. In producing gametes, the male reproductive system must also produce the fluids, ‘foods’ and other important components required for the sperm to be healthy and viable. This fluid is called semen.

During puberty for both sexes, there are many obvious physical changes that occur.

- 1 Describe three physical changes that might occur when a male is passing through puberty.

Sperm originates in paired organs called testes (singular *testis*). Compared with the female gamete (oocyte or ‘egg’) a sperm is tiny. Testes are housed in a muscular sac called the scrotum. Sperm’s streamlined shape is designed to transport one half of the father’s genetic material (DNA) to the egg for fertilisation – as quickly as possible. Therefore it contains a DNA-packed ‘head’, the top half of which is covered by a cap. The cap is made of enzymes (biological catalysts) which will help to dissolve the outer coating of the egg, thereby allowing the DNA to be taken into it. The mid-piece contains mitochondria, organelles whose function is to provide energy for a cell. The sperm’s tail is used to propel it through the female reproductive tract, using mitochondria to produce its energy.

- 2 Define *gamete*.

The testes contain long, wound-up tubes called seminiferous tubules (*seminiferous*, meaning ‘seed-bearing’). Sperm originates in the walls of the seminiferous tubules.

The sperm cells become progressively more mature as they move toward the centre of the tubule. (See figure 1.) By the time the sperm reaches the centre of the tubule, it has a tail with which to propel it but it must still travel up to the nearby epididymis in order to complete maturation. The epididymis is located upon each of the testes (*epi*, meaning ‘upon’ + *didymos*, meaning ‘testicle’).

One of the important hormones produced by the testes is called testosterone. Without it, sperm would not mature and puberty would not begin.

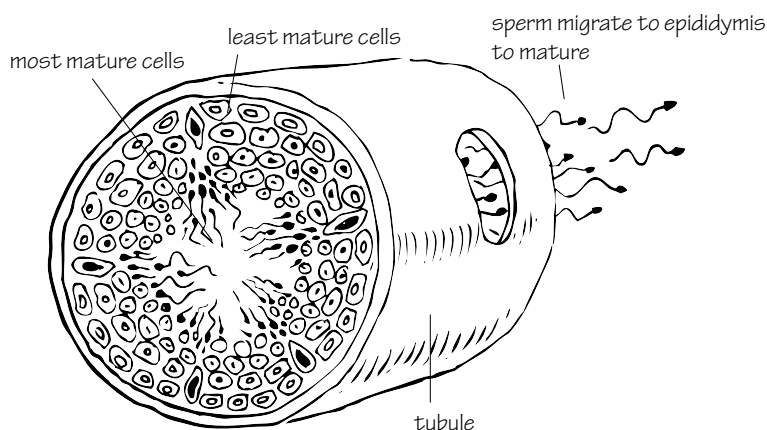


Figure 1

- 3 What does the acronym DNA represent?

- 4 What substances are present in semen?

- 5 Define *enzyme*.

- 6 Which organ houses the testes?

- 7 Define *hormone*.

- 8 Name a hormone produced by the testes and identify the role that it plays.

- 9 What role do mitochondria play in cells?

- 10 The stage where genital organs begin to mature and when secondary sex characteristics start to appear – what does this define?



In order for a sperm to be able to survive and travel, it needs the help of other reproductive organs. Once the mature sperm leaves the epididymis, it journeys along a tube called the urethra which will eventually enable it to leave the body. Along the way, paired seminal vesicles will have produced seminal fluid to help the sperm to survive. Seminal vesicles are the first things to contribute to the sperm's successful journey out of the body. The seminal fluid contains fats, sugars and amino acids. It is released into the urethra as required. The prostate gland also produces fluids that will be mixed with the seminal fluid. The prostate gland is located near the base of the bladder (the bladder is a sac that collects and stores urine). Prostate fluids are designed to protect the sperm against the acidic nature of the vaginal fluids. Think of it as an alkaline 'coat' that surrounds the sperm. Finally, the paired Cowper's gland (also called the bulbourethral gland or Mery's gland) adds its own fluid to the mix. This fluid is designed to increase the amount of lubrication to assist in intercourse.

- 11** Defective sperm can have these faults: bent tail or two tails, two heads, microencephaly (*micro* meaning 'small', *encephalos* meaning 'head'), or macroencephaly (*macro* meaning 'large'). Describe how each fault could create problems with fertilisation.

- 12** Label the sperm cell below (figure 2). Include a brief description of the function of each part.

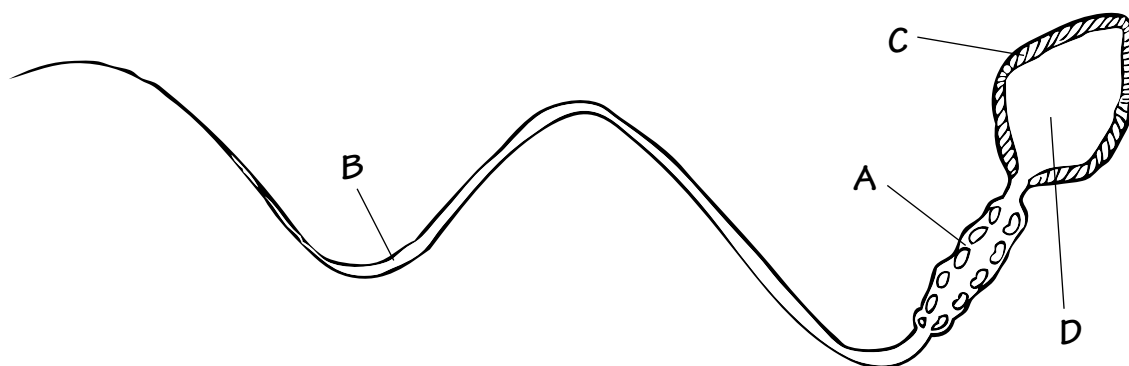


Figure 2

- 13** Sperm are described as motile. What does *motile* mean?
-
- 14** A substance was described as a *spermicide*. What does this mean? _____
- 15** Mark the following statements as true or false. Correct the false statements so that they are true.
- a** Another name for the male gonads is testes. _____
 - b** Semen only contains sperm. _____
 - c** Female gametes are called ova. _____
 - d** Male sex cells are much larger than female sex cells. _____
 - e** All males begin to produce sperm at the same age. _____
 - f** Sperm leave the body through the urethra. _____
 - g** Male DNA cannot enter the egg until enzymes dissolve the egg's outer coating. _____
 - h** The prostate produces an acidic coating for sperm. _____



- 16** On the diagram below (figure 3), label the following and write a brief piece of information about each: *urethra, seminal vesicles, scrotum, prostate gland, seminiferous tubule, Cowper's gland, testis, epididymis.*
- 17**
- a** Draw red arrows to show the direction of sperm flow from its point of origin to exiting the body.
 - b** Highlight the area at which seminal fluid meets and mixes with sperm.
 - c** Draw green arrows to indicate the direction of urine flow through the urethra.
 - d** Draw blue arrows to indicate where seminal fluid joins sperm and exits the body.
 - e** Draw a black arrow to indicate where Cowper's Gland fluid enters the scene.

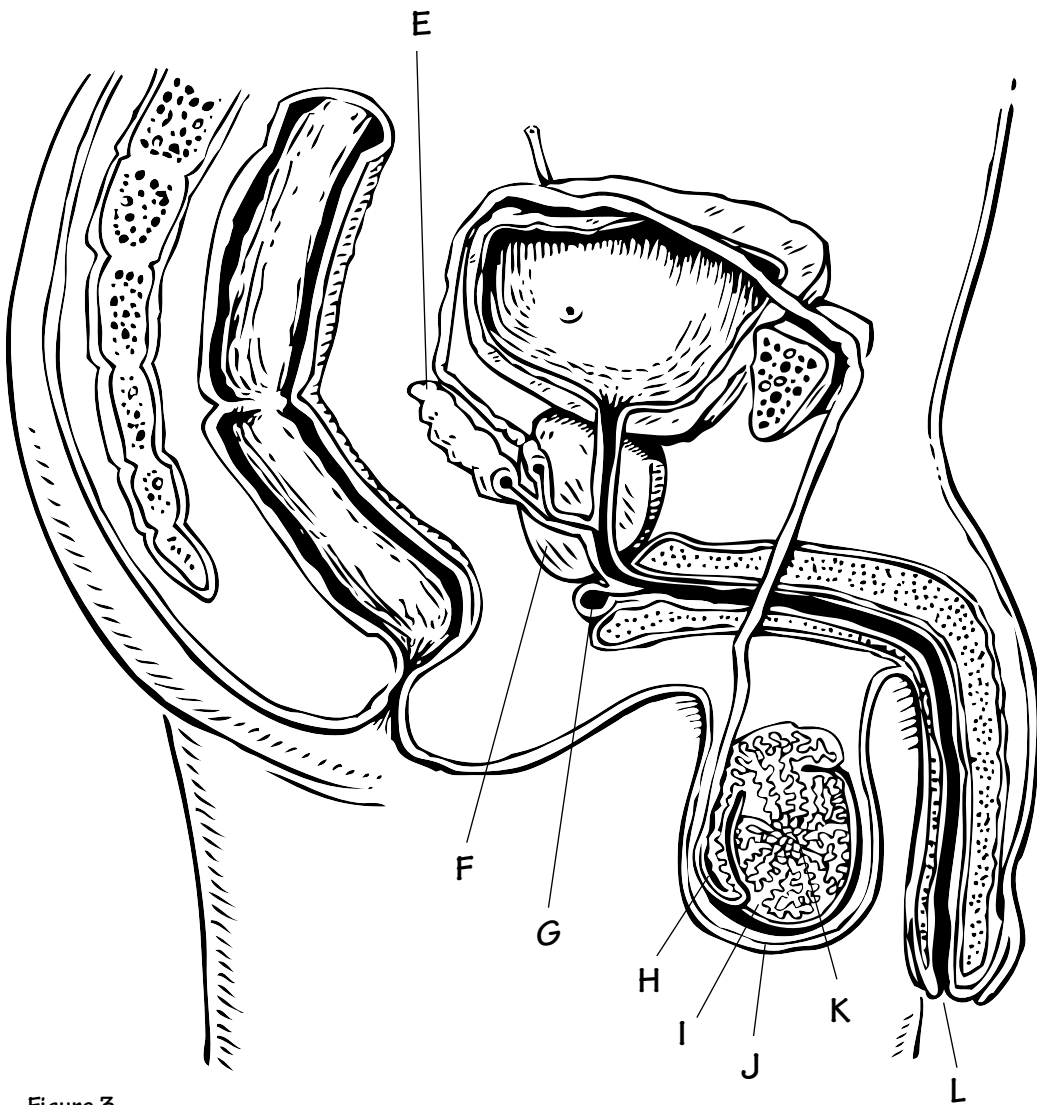
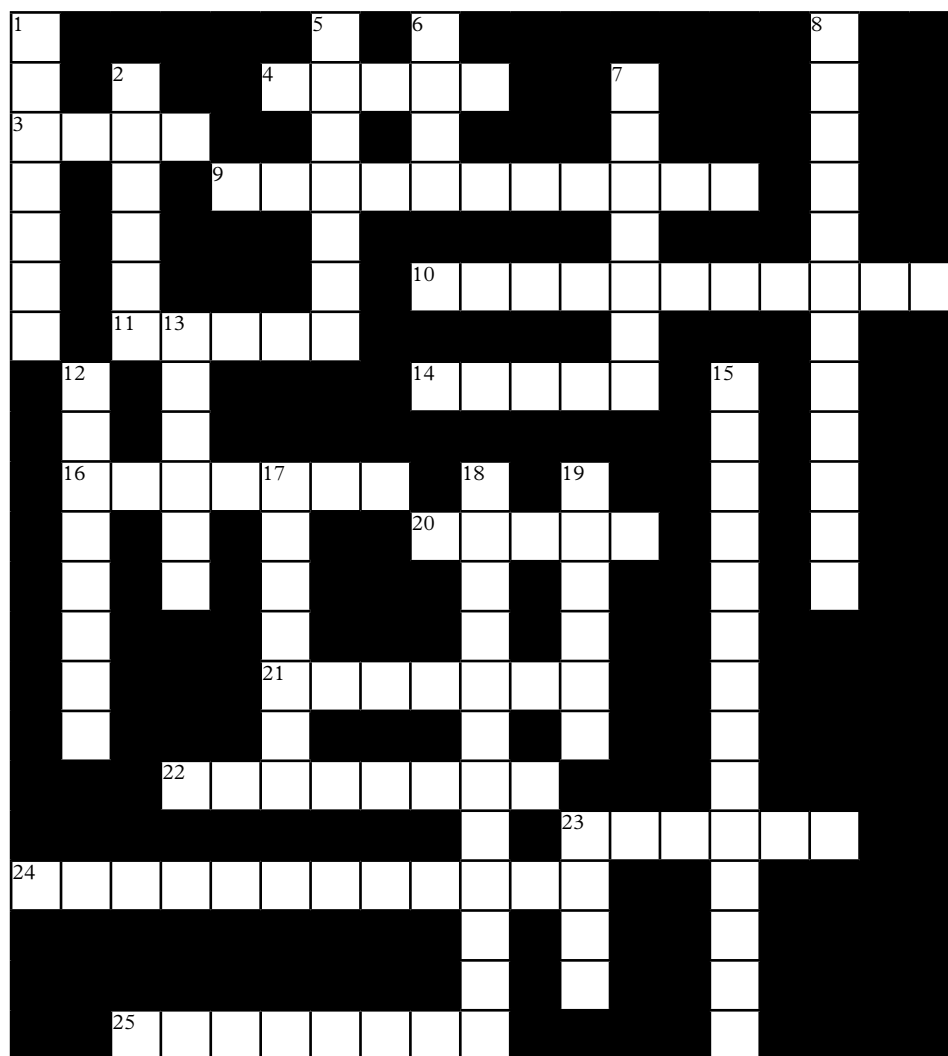


Figure 3



Clues

Down

- 1** Pair of glands that add fluid to semen at the base of the penis
- 2** Developing human in the womb
- 5** Onset of sexual maturity
- 6** Female sex cell
- 7** Biological catalysts
- 8** Tubes producing semen
- 12** Gland that secretes alkaline fluid into semen
- 13** Canal from the uterus to the external part of the female genital organs
- 15** Conduct the egg from the ovary to the uterus
- 17** Carries urine from the bladder
- 18** Male hormone responsible for secondary sexual characteristics
- 19** Pair of male reproductive glands
- 23** Oily or greasy substances

Across

- 3** Also called the uterus
- 4** External part of female genital organs
- 9** Male sex cells
- 10** Mucus membrane lining the uterus
- 11** Where female sex cells are stored
- 14** Major part of the skeleton
- 16** Tube that takes eggs from the ovary
- 20** Fluid containing sperm
- 21** Produced by glands and affects body processes
- 22** Connects foetus to uterus
- 23** Developing human, 3 months after conception
- 24** Energy producers in cells
- 25** Having a pH greater than 7



- 1** What role does seminal fluid play in the reproductive process?

- 2** Explain how puberty differs from childhood.

- 3** Why does the womb need to be sealed during pregnancy?

- 4** Explain why a developing embryo needs hormones if it can ‘share’ its mother’s hormones.

- 5** What are the substances testosterone and oestrogen? How do they differ?

- 6** Compare the function of the urethra in both sexes.

- 7** Describe the difference between an embryo and a foetus.

- 8** Explain why sperm is so much smaller than the egg and why the egg must be so much larger.

- 9** What is amniotic fluid and what is its function?

- 10** Briefly describe the function and process of menstruation.

- 11** Briefly explain what happens once an egg is fertilised.

- 12** Explain why the female cannot use her reproductive system to release urine in the same way that males can release semen and urine via the same route.



The term ‘ethics’ is derived from the Greek word *ethika*, which can be loosely translated as ‘the proper conduct, customs, principles or standards of how humans should behave’.

Ethics is the study of how humans behave, morals, what is right and wrong and what is good and bad.

Stem cells are undifferentiated cells (cells that have not yet committed to developing into a specific tissue type). They are present in developing embryos. They also occur in adults, but in lesser numbers. If stem cells are removed before they follow specific growth patterns, researchers can direct their course of development. It is hoped that stem cell research will one day allow replacement organs to be grown for impaired individuals, and help sufferers of diabetes, Parkinson’s disease and many other conditions.

Embryonic stem cell research is extremely controversial, because the cells can be obtained from aborted fetuses. Many people are concerned by the idea that our society could use ‘crops’ of embryos in order to help or save other people. They think that abortion rates may increase, justified ‘in the name of research’, and some believe that it takes away an embryo’s right to life. Those in favour of embryonic stem cell research point out the benefits to society in helping people who have potentially curable conditions.

- 1 In your workbook, list arguments for and against embryonic stem cell research.
- 2 Many governments have banned the extraction of stem cells from embryos. However, many people support the cloning of existing stem cells (those extracted before the bans) to create ‘lines’ of stem cells. Do you agree or disagree with this? Justify your answer.

- 3 *A cell that can develop into a number of different kinds of cells, depending on the specific chemical signals received – what does this describe?* _____

The umbilical cord is rich in stem cells. Newborn infants no longer need their umbilical cords after they are born. Until recently, umbilical cords were discarded. A number of diseases have been treated using umbilical cord stem cells. These include leukaemia and sickle cell anaemia.

- 4 What is the purpose of the umbilical cord? Explain why a newborn no longer needs it.
- 5 New parents are sometimes asked if they wish to donate their baby’s umbilical cord to medical science. Many people are happy to do this but others refuse. Suggest reasons for and against doing this.

- 6 **Research** – Investigate a disease that might be helped by stem cells and explain how they could help.

- 7 **Research** – Stem cell research is showing promise in the repair of damaged nerves in some quadriplegics. Christopher Reeve (former ‘Superman’ actor) was a quadriplegic as a result of a horse-riding accident. He took part in stem cell research and was hopeful that it had a promising future. What is a *quadriplegic*?

- 8 **Research** – The terms *embryo* and *foetus* are often confused. Define:

a embryo _____

b foetus _____



An organ system is a collection of tissues or organs that work together, each with a specific task (or tasks) related to the healthy functioning of an organism. Plants and animals both have organ systems (Latin *organum*, meaning 'tool, instrument'). Plants have two main organ systems – one works below the ground, the other above ground.

- 1 Name the organ system that works below ground. _____
- 2 Name the organ system that works above ground. _____

Roots, tubers and rhizomes form the plant's underground organ system.

Roots are usually (but not always) found underground. Their function is to absorb water and nutrients for the plant's growth and development. Roots also store excess sugars and act as anchors, holding the plant in place and giving it stability as it grows. Tiny root hairs grow off roots to increase the roots' surface area.

Tubers act like a pantry. They store food for the plant. Buds are often present on tubers (*tuber* comes from a Latin word meaning 'swelling' or 'knob').

Rhizomes are underground stems that grow horizontally (creep along) underground. Rhizomes grow roots and also store food, allowing a plant to spread across an area. Shoots, leaves, flowers and fruits are examples of a plant's above-ground organs.

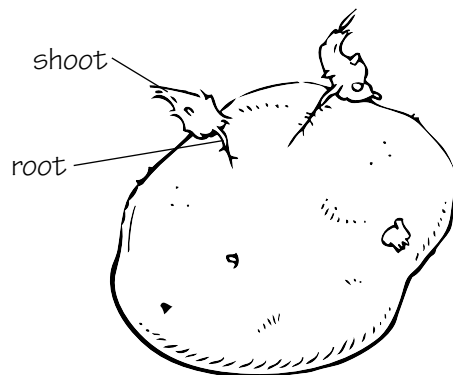


Figure 1

- 3 Would a potato be considered a tuber? Explain. _____
- 4 Name a tuber that is not a potato. _____
- 5 Name two ways in which weeds invade areas. _____
- 6 Would a rhizome be considered a modified plant stem? _____
- 7 Mark each of the following statements as true or false. Then correct the false statements so that they are true.
 - a Root hairs increase the amount of water and nutrients plants can absorb. _____
 - b Tubers are often edible. _____
 - c A leaf is not an organ. _____
 - d Some root systems grow above ground. _____
 - e Plants cannot store excess sugars. _____
 - f Tubers are a means of plant reproduction. _____
 - g Plants arising from tubers are genetically identical to the parent plant. _____

Plant tissue

Plants have three kinds of tissue: dermal, ground and vascular.

Dermal tissue acts like an outer 'skin' on a plant. It covers the outer parts of herbaceous plants. Herbaceous plants are 'non-woody' plants (think herbs). Dermal tissue helps to protect the plant. However, it is not sufficiently insulating to protect against frost. For this reason, the ground tissues of herbaceous plants die in cold weather but the organ system below ground survives. This allows the plant to sprout and grow again when temperatures rise.

- 8 List an example of a herbaceous plant. _____
- 9 List an example of a woody plant. _____

Ground tissue makes up the bulk of the plant's tissues. It is all the tissue excluding vascular tissue and the outer, protective, dermal tissue. Vascular tissue is the transport tissue (Latin *vasculum*, meaning 'small vessel').

Vascular tissue is made up of xylem and phloem. Xylem transports dissolved nutrients (minerals and ions) and water up from the roots to the leaves. The plant uses these things, along with sunlight and carbon dioxide (CO₂) to make its own food. When the food is made, phloem is used to transport it throughout the plant.



In summary, xylem transports raw material upward from the roots, phloem transports the manufactured foods throughout the plant. Figure 2 shows a stem cross section of a typical vascular plant.

- 10** In your own words, describe the difference between *xylem* and *phloem*.

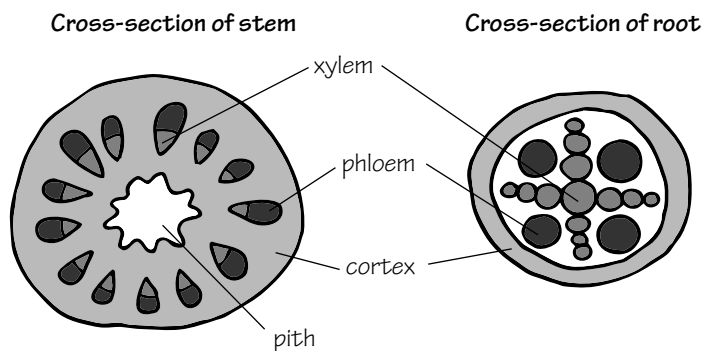


Figure 2

Plant groups – a summary

Non-vascular plants

Bryophytes are the simplest of the true plants. Mosses, liverworts and hornworts are bryophytes, though some sources restrict bryophytes to mosses only (Greek *bryon*, meaning ‘moss’ + *phyte/phyton*, meaning ‘plant’). Scientists believe that bryophytes evolved from algae, because they pass through stages during their growth that are similar to algal development. This idea gains further support from the fact that many species can be found in water.

Bryophytes exist in most environments except those that are extremely hot or extremely cold. However, they prefer moist, warm tropical or sub-tropical areas. They are small, low growing plants that most people will walk right past without even noticing. Bryophytes cannot grow large because they have no vascular system. They also lack a true root system.

- 11** Where in Australia would you expect to find bryophytes? Explain your answer. _____

Mosses form the larger of the bryophyte groups. They are soft, velvety, delicate plants that are quite beautiful on close inspection. Liverworts resemble mosses in many ways. The main difference between the two is their means of reproduction.

Hornworts resemble liverworts. The name means ‘horned liverwort’ and comes from the long capsules on the plant that look a bit like little horns. Hornworts prefer still (or slow moving) waters but will grow in other damp places.

- 12** Bryophytes do not have true roots. If they do not have a vascular system, would their ‘leaves’ be considered true leaves? Explain.

- 13** Suggest how bryophytes might prevent erosion. _____

- 14** It is always best not to trample plants, particularly so with bryophytes. Why? _____

- 15** Why are bryophytes restricted to moist environments? _____

- 16** **Research** – Algae are not plants but it has been said that algae are as important in water as grasses are on land. Discuss why both are considered important to the places in which they are found.



Vascular plants

Vascular refers to ‘vessels’ and vascular plants are those which contain vessels used to transport materials. (Think of them as the vein network of plants.) Plants could quite happily live close to the ground, but they need sunlight to survive. Without it, photosynthesis cannot occur. The best way for plants to be able to reach up towards the sunlight is to develop a vascular system and leaves. These enable plants to grow taller and ‘reach’ for sunlight from more awkward places where they might otherwise be shaded.

- 17** List an example of a vascular plant. _____

Vascular plants can be found with or without seeds. Ferns, horsetails and lycopods are seedless vascular plants. These plants are often called ‘living fossils’ because they have been around since prehistoric times. Ferns were the least dominant of the three types in the past, now they form the largest group.

The major vascular plant group is the seed plants. There are several reasons why they have become the dominant plant on Earth. Their root systems grow much deeper underground, allowing them to absorb water that is harder to reach, and anchoring them more effectively to the ground. Their vascular systems are much more efficient at transporting nutrients and their large surface areas of leaves photosynthesise more effectively.

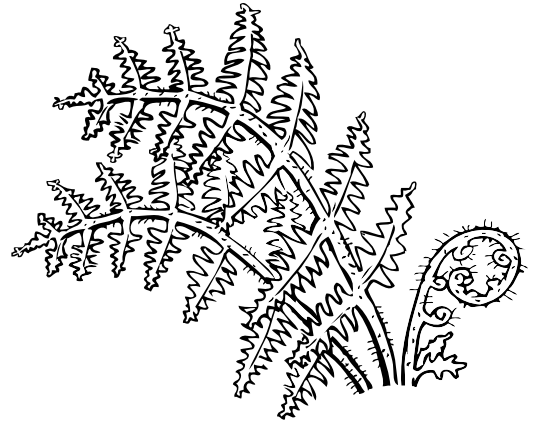


Figure 3

- 18** What type of plant is shown in figure 3? _____

Hardwood vs softwood

You may be familiar with the terms ‘hardwood’ and ‘softwood’. The names do not refer to the hardness of the tree’s wood, but to the seeds they produce (although most fit the description well in terms of the density of their wood).

Hardwoods are called angiosperms or flowering plants. Angiosperms produce seeds that are covered with a natural, protective coating (some coatings are harder than others). The coats are true fruits produced by flowers. Angiosperms represent two-thirds of the land plants. The name comes from *angio*, a word meaning ‘seed vessel’.

Softwoods are called gymnosperms. Gymnosperms are the antiques of the seed plants and many of the species are now extinct. The word comes from the Greek *gymnospermnos* meaning ‘naked seed’. Gymnosperms produce seeds with no hard or protective covering. They produce coatings around their seeds, but these tend to be cones (woody or soft flesh) and not true fruits. Each cone holds many seeds. Once the cone opens, the seeds fall to the ground. They have no outer coating to protect them until they germinate.

Angiosperms and gymnosperms are now often lumped together as spermatophytes.

- 19** List three examples of angiosperms.

- 20** Would a pine tree be considered a hardwood or a softwood? Explain your answer.

In order for a viable seed to be formed, a male sex cell must fertilise a female cell.

- 21** What is the name of the male sex cell of a plant? _____
- 22** What is the name of the female sex cell of a plant? _____
- 23** For many years, humans have collected seeds. List three reasons for this.



The word *transpiration* comes from the Latin *trans* meaning ‘across’ or ‘through’ and *spirare* which means ‘to breathe or pass something through pores or tissue’. Transpiration is the word used to describe the loss of water from a plant, in the form of water vapour. The water is lost through tiny pores in the leaves. These pores are called stomata. The singular form of stomata is *stoma*, meaning ‘mouth’. Under the microscope, a stoma looks a bit like a little mouth, with lips that open and shut. Usually stomata open at sunrise and close at night. (Although some desert plants have stomata that open at night.) Therefore stomata, like the chloroplasts described below, are also activated to a large degree by sunlight.

- 1 Write another word for *vapour*. _____

Throughout the day, stomata open and close as the plant requires. When the plant is low on water, they close to prevent further water loss, although a certain amount is still transpired. When water is readily available, stomata open. In average conditions, over 90% of the water a plant takes up through its roots is transpired.

All vascular plants transpire. *Vascular* means ‘vessel’. Vascular plants carry water and nutrients (dissolved minerals) through a system of vessels or ‘tubes’. The vascular system is easily observed by cutting a stick of celery crossways. To see transpired water vapour, tie a clear plastic bag over the cut section and leave it for several hours to become visible. You should see the colour drawing up the celery. If left for long enough, the leaves will also start taking in the colour.

Plants actively photosynthesise during periods of light. This is when transpiration action is at its greatest. It is important not to confuse *transpiration* with *evaporation* – they are two different processes.

- 2 Humans perspire (sweat). Is perspiration a form of transpiration? Explain. _____
- 3 Animals also have vascular systems. Name one animal vascular system. What does it carry? _____
- 4 Briefly describe the differences between transpiration and evaporation. _____

How plants obtain food

Like all organisms, plants need food. Plants are autotrophs, which means they manufacture their own food. Plants need foods such as starches, sugars, proteins, dissolved minerals and water. Once the food is manufactured, it must be distributed throughout the plant because not all plant cells can make their own food. Three important parts of the plant take care of this: chloroplasts, xylem and phloem.

- 5 Animals are *heterotrophs*. What does this mean? _____

Plastids are tiny organelles. They have the name ‘organelle’ because they can be compared to the organs in our bodies. Like our organs, organelles each perform specific tasks within a cell.

Chloroplasts are special plastids that contain a green pigment called chlorophyll. *Chloros* means ‘green’ and *phyll* means ‘leaf’. Wherever a plant is green, you will find chloroplasts in the cells of the plant tissue. In the presence of light, chloroplasts are activated. They convert carbon dioxide and water into glucose. Cricket players often have to remove crushed chloroplasts from the knees of their trousers!

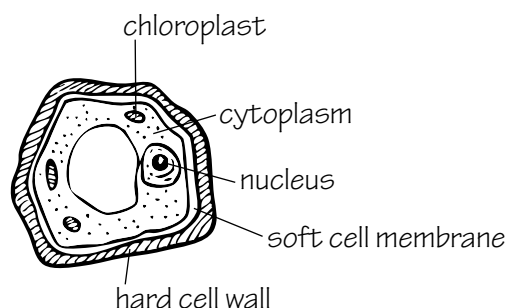


Figure 1

- 6 *Xantho* means ‘yellow/orange’. What colour would you expect a plant containing xanthophyll to be? List an example of a plant that contains xanthophyll. _____

Try this! The next time you put cut flowers in a vase, add some food colouring and watch how the flower petals change as they take up the coloured water.



Xylem and phloem

Many students find it difficult to remember the functions of, and differences between, xylem and phloem. If you do, you will be pleased to learn that it is not as confusing as it may seem! Plants require water and nutrients to survive. They convert the raw materials of water, dissolved nutrients and energy from sunlight into carbohydrates and other food they require to survive. Leaves are the part of a plant that manufactures the majority of food.

Xylem (pronounced *zye-lem*) is the woody tissue of a root or stem that carries raw materials, such as water and inorganic nutrients, up into the plant from the ground. The plant uses them (with carbon dioxide from the air) to make food. Roots contain many tiny root hairs which increase the surface area available to draw the raw materials up into the plant. Xylem also provides structural support. Wood is made of xylem.

Phloem (pronounced *flow-em*) can be considered the opposite of xylem. Phloem carries water and organic substances away from the leaves to the rest of the plant. Once the leaves have manufactured food from the raw materials the xylem delivered, that food is shared around the plant to nourish its cells. Phloem's job is to deliver it.

Hint! A phonetic trick may help you to remember the difference between xylem and phloem. Phloem distributes the manufactured foods. Remember, 'phloem follows with food'. Phloem can't follow until xylem first delivers the raw materials to the leaves.

Cohesion and adhesion – sticking together – You might wonder how water travels up a huge tree when it would seem that gravity would be pulling it down. Water is a polar molecule. This means that it has a slightly negative end (δ^-) and a slightly positive end (δ^+). Polarity tends to attract opposite charges, which causes water molecules to cling to each other or to other molecules. *Cohesion* describes the force of attraction between molecules that are the same type, such as water molecules. Every water molecule has the same H_2O structure and composition, so when they attract each other they are cohering. *Adhesion* describes a similar attraction, but the attraction is between unlike molecules. You will have heard of 'adhesive tape'. This is a substance that sticks (adheres) to other substances, such as paper.

The vascular system of plants is full of water, just as your blood vascular system is full of blood. The water molecules are attracted to each other because of their polarity. However, the water molecules are also adhering to the xylem tissue. In effect, they use this ability to adhere to other substances to 'climb' up the plant. The water molecules will also contain dissolved nutrients which are adhering to them. This is how xylem delivers the raw materials to the leaves.

The leaves are the 'end of the road'. They manufacture the food, then send it throughout the plant, also using the cohesion–adhesion process. As the stomata release water vapour through transpiration, some space is freed up in the xylem. The water further down then climbs to the top (by adhering to the walls of the xylem). Down at the root level, more water is drawn up to replace it. Think of drawing liquid up through a straw.

- 7 Figure 2 is a concept layer map showing how various concepts are linked in vascular plants. Write in each box to complete the relationships.

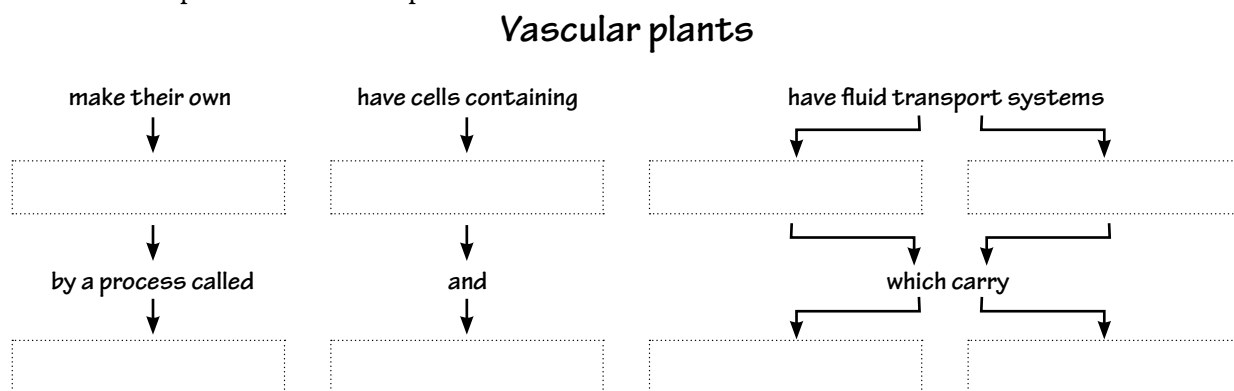


Figure 2

Turgor pressure – The word *turgid* means 'swollen' or 'distended' (*turgere*, meaning 'to swell'). Plant cells have cell walls (unlike animal cells). When the plant cell is as full of water as it can be, it is stiff and the plant stands upright. When turgor pressure is low, such as on hot dry days, the cells lose water and the plant wilts. The cell wall allows the cell to withstand the pressure of being full of water. Otherwise it would burst.

- 8 Why do people place limp celery in water? _____



Multiple choice – circle the correct answer.

- 1 When a plant cell holds as much water as it can, it is said to be under a high:
a turbid pressure. **b** turgor pressure. **c** turmoil pressure. **d** turpeth pressure.
- 2 Chloroplasts are specialised plastids that contain which colour pigment?
a yellow **b** red **c** green **d** brown
- 3 The release of water vapour from a plant is called:
a evaporation **b** sublimation **c** condensation **d** transpiration
- 4 Tiny pores that release water vapour from leaves are called:
a stomata **b** stomach **c** stroma **d** stomatic
- 5 An example of xylem tissue is:
a leaves **b** wood **c** roots **d** root hairs
- 6 Which of the following ripe fruits (including skin) would **not** contain chlorophyll?
a watermelon **b** orange **c** celery **d** rockmelon

True or false?

- 7 Many species of seaweed contain chlorophyll. _____
- 8 Plants manufacture their food during the day. _____
- 9 A leaf is an example of a vascular system. _____
- 10 Cell walls are important in preventing plant cells from rupturing. _____
- 11 Transpiration occurs equally during night and day. _____
- 12 Plant and animal cells contain plastids. _____

Extended answers

- 13 Briefly discuss the roles of *xylem* and *phloem*.

- 14 What is the function of root hairs?

- 15 Explain the difference between *cohesion* and *adhesion*.



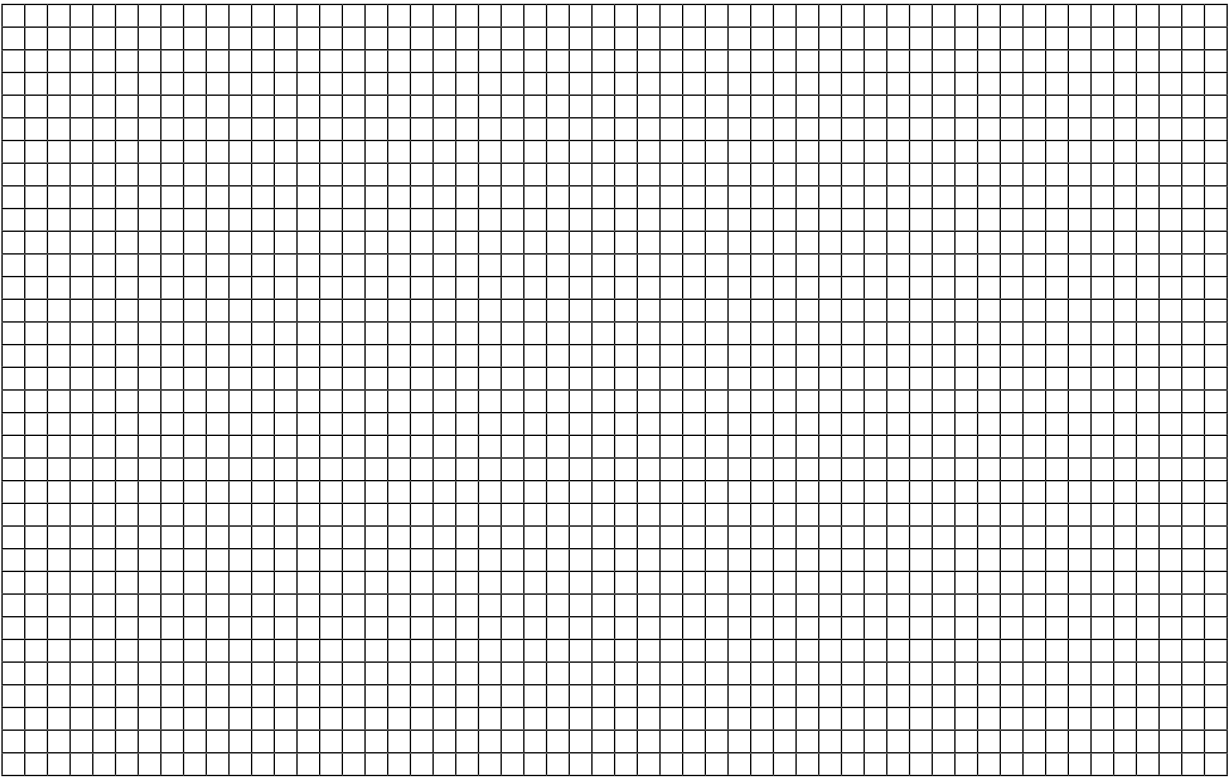
All plants need to take in water. Water is taken in by roots and carried through the plant by tubes called *xylem*. Water is carried to leaves where some evaporates through stomata (pores). This process is called *transpiration*.

Researchers carried out an investigation to measure the transpiration rate (in litres per hour) of a particular tree in Crookwell in the Southern Tablelands of New South Wales. The table below shows the results.

Time (hours)	0	3	6	9	12	15	18	21	24
Rate (L. hr ⁻¹)	1.0	2.1	4.0	6.4	7.8	7.7	4.2	2.2	1.3

Time (hours)	27	30	33	36	39	42	45	48	51
Rate (L. hr ⁻¹)	1.2	3.9	7.4	8.1	7.6	4.1	0.8	0.3	0.9

1 Plot a line graph of transpiration rate (*y*-axis) against time (*x*-axis). Label the axes and show the units used.



- 2 On the graph, show when it is day and when it is night.
- 3 Explain your answer to question 2. _____
- 4 This experiment was carried out in mid-summer. What results would you expect if this experiment was repeated in mid-winter in the same location? _____
- 5 How could you estimate total water transpired per day? _____
- 6 Using the method outlined in your answer to question 5, calculate how much water transpired in the first 24 hours. _____



Some soils in Australia are acidic. Their pH is too low for optimum growth of several types of seeds. To combat this, seeds can be ‘pelleted’; that is, they are covered in a coat containing gypsum or lime. These two compounds contain calcium (Ca) and are basic (pH greater than 7), and so can counteract the effect of acidic soils.

A researcher investigated the effects of pelleting on a particular species of plant seed. Various characteristics were measured 4 weeks after the appearance of a seedling.

Treatment	Height of plant (cm)	No. of leaves per plant	Mass of plant (g)
Control	10.8	13	6.1
Gypsum	13.1	19	14.3
Lime	15.4	27	21.5

- 1 What is meant by ‘optimum’?_____
- 2 Name an element present in both gypsum and lime._____
- 3 The use of a control is essential when carrying out experiments to compare the effects of different things (variables). Explain what is meant by a ‘control’.

- 4 Outline what the researcher should have done to ensure that the test was fair._____

- 5 What is the mean (average) height of the plants? Show your working._____

- 6 What is the mean number of leaves per plant? Show your working._____

- 7 What is the mean mass of the plants? Show your working._____

- 8 Calculate the percentage improvement in growth due to pelleting by completing the table.

Treatment	Increase in height of plant (%)	Increase in no. of leaves per plant (%)	Increase in mass of plant (%)
Gypsum			
Lime			

- 9 Mark the following statements as true or false. Then correct the false statements so that they are true.
a Acidic soils have a pH less than 7._____
b All seeds need to be pelleted to grow well in Australian soils._____
- 10 What alternatives might be available to a farmer who wishes to plant seeds in soil that is more acidic than the optimum for her chosen crop?

- 11 **Research** – Hydrangeas can grow pink or blue flowers, depending on the acidity of the soil. Find out which soil type produces each colour.

1 Read the following passage and explain what is meant by each term in **bold** (listed below).

Australian conditions are not always ideal for the growth of **seedlings**. There can be **prolonged** drought, high temperatures and bushfires. In order for plants to **germinate** at the 'correct' time, i.e. have the greatest chance for the seeds to eventually grow into plants, many seeds remain **dormant** until special conditions occur. The mechanisms that **inhibit** seed from germination can be a chemical present or a thick seed coat. Often people who wish to germinate native Australian seeds have to **replicate** the conditions of the plant's natural **habitat**.

Some seeds have a hard seed coat and this needs to be **abraded** using a substance such as sandpaper. Seeds with their coats treated in this way germinate more quickly than untreated seeds.

Seeds (such as members of the *Acacia* **species**) can also be treated by **immersion** in boiling water. Many seeds from areas that are prone to grass fires may need to experience direct **exposure** to fire and smoke. Fire-activated seeds are particularly interesting. A fire must reach a certain intensity before the seed coat will burst open. The fire that achieves this intensity will also have been severe enough to destroy the parent plant. This strategy prevents seedlings from competing with parents. It also ensures that the species will survive after a severe fire.

The chemicals preventing germination can be removed by a process called **leaching**. Here the seeds are suspended in a solution and the chemicals are gradually extracted from the seeds.

- a** seedlings_____
- b** prolonged_____
- c** germinate_____
- d** dormant_____
- e** inhibit_____
- f** replicate_____
- g** habitat_____
- h** abraded_____
- i** species_____
- j** immersion_____
- k** exposure_____
- l** leaching_____

2 Research – Name three specific Australian plants that need pre-treatment before germination. State the type of treatment each plant requires.

3 Australia is a drought-prone country, yet many people in drought-affected areas are able, to varying degrees, to grow attractive flowering gardens. Suggest how (if watering from taps is not allowed) people manage this.



- 1** Read the following passage and write meanings for the terms in *italics* (listed below).

Asexual reproduction involves only one parent. The parent duplicates its own genetic material, followed by the rest of the cell contents. The cell then divides to produce *offspring* known as daughter cells, which are genetically identical to the parent. Clones can be defined as plants or animals grown from a single cell or plants grown from a single *multicellular* plant. Clones can also be defined as a group of *organisms* that are genetically identical.

Cloning can be used by humans to grow plants when it is difficult to get plants to grow 'normally', for example, when seeds are difficult to obtain, when mass production of genetically identical plants is desired and to increase numbers of a plant threatened by *extinction*. The Wollemi Pine is an example of a plant that exists only in *minute* numbers. Thought to have died out millions of years ago, a small number of *specimens* were discovered in 1994 and since then scientists have been trying to increase its numbers. Seeds have been collected, but only a very small percentage of these have proved *viable*. Scientists have also tried to increase the numbers of Wollemi Pine by taking cuttings and by using *tissue* culture. In tissue culture, a small piece of tissue is taken from a growing part of the plant (e.g. a bud or root). This piece of tissue is then *cultured* in a medium containing the required *nutrients*. The cells in the tissue *replicate* and a group of cells is formed. These are treated with plant hormones and develop into seedlings. These seedlings are *clones* of the original parent. They can then be grown into plants.



- a** offspring _____
- b** multicellular _____
- c** organisms _____
- d** extinction _____
- e** minute _____
- f** specimens _____
- g** viable _____
- h** tissue _____
- i** cultured _____
- j** nutrients _____
- k** replicate _____
- l** clones _____
- 2** Name one advantage and one disadvantage of a cell being able to clone itself. _____
- _____
- 3** Outline the difference between cloning and inbreeding. _____
- _____
- _____
- 4** Taking cuttings is one way of vegetative propagation that produces clones. Number the following steps of propagation in the correct order, from first (1) to last (8).
- | | |
|--|---|
| a Clones multiply | e Cutting grows larger |
| b Shoots are cut from plant | f Cutting is washed |
| c Plant growth hormones are added | g Cutting is transferred to growth medium |
| d Cutting is trimmed of excess foliage (leaves) | h Developing plant is transferred to larger growing area |



On a Norwegian island 1000 kilometres from the North Pole, a consortium of nations is beginning work on a super-safe repository for 'the world's most precious asset': its seeds.

When the chilled vault is completely installed in the permafrost and above the tsunami line some time next year, more than 1 million samples of agricultural seed, from apples to Zinfandel grapes, will be brought to the 'seed bank of last resort'. Yesterday the prime ministers of five countries and a small throng of other officials were set to lay the cornerstone for what will be, in effect, the Fort Knox of seeds.

The vault, 50 metres long and five metres wide, is to be bored into the

side of a mountain on the island of Spitsbergen.

'It will be lined with a metre of the hardest prestressed concrete,' said Cary Fowler, executive director of the Rome-based Global Crop Diversity Trust, which will manage the repository. The vault above the village of Longyearbyen will stand higher than any projected rise in sea levels, even if a tsunami strikes.

If the power fails, scientists will have months to fix the electric cooling system before the vault gets too warm. That is because the temperature on Spitsbergen, part of the Svalbard archipelago far above the Arctic Circle, seldom goes higher than minus 3 degrees.

Even if it gets above freezing, the

vault would be insulated by the mountain and its concrete walls 'for months and even years', Dr Fowler said. 'Neither nuclear war nor global warming nor virtually any other imaginable threat will disturb the seeds in their aluminum packets,' Dr Fowler said.

Seed samples being stored around the world were 'the world's most precious asset', he said. 'It is beyond value, and I don't think anyone can imagine realistic solutions to problems like climate change, water shortage or energy shortage without a system of agriculture that can adapt and change.'

Jeff Nesmith, *The Washington Post/Sydney Morning Herald*, 19/06/05

1 Write meanings for the following terms.

- a** consortium _____
- b** permafrost _____
- c** tsunami _____
- d** vault _____
- e** archipelago _____
- f** global warming _____
- g** asset _____
- h** agriculture _____
- i** repository _____
- j** diversity _____
- k** climate _____

2 Assuming that the vault is as wide as it is high, what volume of rock needs to be bored out? Show your calculations.

3 Outline why aluminium is a good material to use for making packets for seeds.

4 Suggest why some people think it is worthwhile to store seeds in this way.



5 Research – Why is ‘Fort Knox’ mentioned in the article?

6 The advantages of having a seed vault prepared in case of disaster are evident. Suggest what difficulties might be faced if we lost all of our crops and needed to open the stored packages.

7 If wide-scale nuclear war were the reason for crop destruction, what difficulties might the survivors face?

8 The seeds stored are called the world’s most precious asset. Do you agree? If so, why? If not, what do you consider to be the world’s most precious asset? Why?

9 If you had to choose three kinds of seeds to be preserved, which ones would you choose? Why do you think these three kinds of seeds are the most important?

10 Imagine you are part of the team who is re-establishing food supplies after a major disaster. List ten things that would need to be done, in order, to get the food grown, harvested and distributed. Include an estimated time frame for each step.



Biodiversity refers to the total range of organisms (including their genetic information) in a particular habitat. Most scientists believe that biodiversity is essential for human life to survive on Earth. The greater the range of species in an ecosystem, the greater the chance that this ecosystem will survive sudden changes such as fire or drought. Grasslands that have many different types of flora recover more quickly from grazing than similar areas with just a few species of plants.

In the 1840s, a large proportion of Ireland's population was dependent on potatoes for food. Almost the entire potato crop then came from just two varieties. In 1845, a fungus (*Phytophthora infestans*) arrived in Ireland from North America. This fungus infected the Irish potato crop causing a blight (type of disease) that destroyed the potatoes. The next four years saw the continuing failure of the potato harvest due to blight. It is estimated that over a million people died of starvation or diseases related to the famine that resulted.

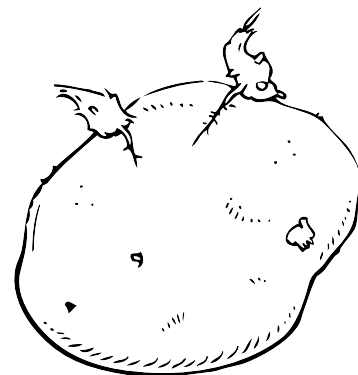


Figure 1

Many scientists today are worried about farmers using only one or two varieties in their crops or livestock. They are concerned that a disease could wipe out a region's entire production of food if there is a reliance on a few types of organisms, instead of a wide range. People raising plants and animals as a business are under great pressure to grow these as efficiently as possible. There is a temptation to concentrate only on a single breed which may grow more quickly or give a greater yield than other breeds.

The clearing of rainforests, bush and other natural habitats is reducing biodiversity. Many wild plants (and animals) are in danger of becoming extinct. It is becoming more common for seed banks to be set up by geneticists to ensure the conservation of some endangered species. Seed banks store seeds at low temperatures so they remain viable for many years. This is an expensive process and it is very difficult to ensure that all species are represented.

1 Write meanings for the following terms. (You may need to refer to a textbook.)

- a** biodiversity _____
- b** habitat _____
- c** ecosystem _____
- d** species _____
- e** flora _____
- f** crop _____
- g** extinct _____
- h** conservation _____
- i** endangered _____
- j** viable _____

2 You are a potato farmer currently growing four varieties for different markets. You are approached by a large food chain that offers you more money to concentrate on a single type of potato – the one that is easiest to grow. Comment on the possible advantages and disadvantages of this new strategy.



Selected answers



1 – The human heart

- 1 The beginning of a human in the womb
- 2 Top
- 3 b
- 4 e.g. Kidney
- 5 Yes, blood is that
- 6 **a** True **b** False **c** False **d** True **e** True **f** False **g** True **h** False
- 7 **a** Aortic valves
b Pulmonary valves
- 8 Stethoscope
- 9 Can't be repaired
- 10 Heart needs it because it is the organ that delivers it throughout the body, brain needs it or cell death occurs
- 11 They might survive but would have irreversible brain damage if they did
- 12 Need to push blood harder
- 13 CO₂
- 14 Fluid-filled spaces
- 15 **a** Heartbeat **b** sInoatrial **c** oxyGenated **d** Heart **e** Wall **f** Atria **g** gravitY **h** HIGHWAY **i** Circulatory
- 16 Blood flow from right to left
- 17 Greater pressure, greater flow of blood
- 18 Some of the blood meant to be moving forward has fallen back
- 19 e.g. Car engine, water supply
- 20 Teacher to check
- 21 Blood flows into RA from body, then RV. Out into lungs and back into LA then LV to body, etc.

2 – Blood & blood flow circuits

- 1 Push further/harder
- 2 Push further/harder
- 3 Blood flows into RA from body, then RV. Out into lungs and back into LA then LV to body, etc.
- 4 A sound-magnifying disk is attached to two tubes, with earpieces at the ends, through which the physician listens
- 5 Not necessarily, it may be normal for some people, 120/80 is a standard that varies with size, age, etc.
- 6 Heart isn't getting sufficient rest with a reading of 100 and is contracting a bit harder than is recommended for good health
- 7 **a** True **b** True **c** False **d** True **e** True **f** False **g** True **h** True
- 8 Increased and long term excessive pressure on the arteries. Heart isn't getting sufficient rest between beats
- 9 Can lead to stroke, heart attack, heart failure, very dangerous
- 10 Patient's blood pressure lower than acceptable (Hypotension)

11 Dizziness, fatigue, nausea, blurred vision, clammy skin

12 Both dangerous, usually high blood pressure is considered more dangerous but depends on degree of condition

13 Damage to heart, manifested as crushing pain behind breastbone. Heart attack leaves permanent damage/death of heart tissue, angina does not

3 – Gases in blood

- 1 **a** Plasma **b** Antibody **c** Red blood cells **d** Oxygen **e** Carbon dioxide **f** Haemoglobin
- 2 **a** Can dissolve no more **b** Study closely **c** Relating to the artery **d** Movement from high to low concentration **e** Thin sheet **f** Allows particles to go through it **g** Describes hydrogen ion concentration **h** Living
- 3 **a** True **b** False **c** False **d** False **e** True (without cells) **f** True **g** True

4 – Major organs – the liver

- 1 Poison
- 2 Teacher to check
- 3 Make into an emulsion, break into droplets that don't form solution
- 4 Stones, hard deposits in the gallbladder caused by excess cholesterol preventing bile from reaching the gallbladder
- 5 'Loose' nature of the strung cells.
- 6 Detoxification to remove toxins, metabolic activities to do with metabolism and other biological and chemical activities
- 7 Painkiller, for minor pains
- 8 To avoid overdoses
- 9 No, if there is no instructed rise in body temperature (from the hypothalamus) it will remain as it is. It reduces temperature by countering the brain's instructions to raise it
- 10 Less stomach irritation, can use with young children
- 11 Teacher to check, better not to get sick in the first place
- 12 Carbon, hydrogen and oxygen
- 13 Teacher to check, Water retention (oedema and ascites), itching caused by bile deposits in skin, toxins to brain, blood, high blood pressure, sensitivity to medications, etc.
- 14 Teacher to check; e.g. Blood transfusion or other forms of sharing blood, sharing infected syringes, tattoos, sexual transmission and prenatal, latter two least likely
- 15 **a** True **b** False **c** True **d** False **e** True **f** True **g** False **h** True **i** True **j** False **k** True **l** True **m** True **n** False

5 – The kidneys

- 1 Artery
- 2 Vein
- 3 Thin, long vessel
- 4 It is a good transport and dissolving medium
- 5 Sac
- 6 Close to
- 7 Coiled
- 8 Glucose, amino acids, water, minerals
- 9 Away from the point of attachment
- 10 Urine
- 11–12 Teacher to check
- 13 To keep the blood clean
- 14 **a** False **b** True **c** True **d** False **e** False **f** False **g** True **h** True **i** False **j** True
- 15 An endocrine gland
- 16 A substance that increases urine output from the kidney. Water, tea etc. are diuretics. Drugs also exist
- 17 Poisonous
- 18 Filtrate is a substance that has passed through a filter; e.g. tea seeped from bags
- 19 Bladder
- 20 Yes it is so long as the remaining one is healthy
- 21 Excretion
- 22 e.g. Trauma, drug or alcohol abuse, secondary effects of infectious diseases

6 – The endocrine system

- 1 Control the body
- 2 Fast vs slow, electrical impulses vs chemicals
- 3 Reproduction, growth and development, control of the internal environment and regulation of energy production
- 4 **a** Too little **b** Too much
- 5 Endocrine
- 6 Endocrine glands
- 7 Hypothalamus
- 8 Carbon, hydrogen, oxygen and nitrogen
- 9 Via the bloodstream
- 10 Heart beat faster, increases breathing rate and also diverts blood supply to muscles
- 11 Two
- 12 Above the kidneys
- 13 Body getting out of step with hormonal rhythms
- 14 Contains vocal cords
- 15 Gullet
- 16 Metabolism
- 17 Thyroid Stimulating Hormone; Pituitary



Selected answers (cont'd)



- 18 Controls so many things
19 Manages calcium levels in the blood, regulates lipids and carbohydrates
20 Yes, it would still be overactive; there would just be less output so the overflow of hormone is slowed
21 This is often the last resort so is useful because it must be treated
22 It's very hard to destroy the exact amount to correct the balance so patient requires HRT for the rest of their life
23 Pancreas
24 Dietary restrictions and/or injections of insulin
25 Womb
26 Stimulating ovulation and preparing the uterus (womb) for pregnancy. The development of secondary sexual characteristics – beard growth, deep voice and muscle, breast development
27 **a** True **b** False **c** True **d** True **e** False **f** True **g** True **h** True **i** False **j** False **k** True **l** False **m** True **n** True **o** True **p** True **q** False

7 – Are you cut out for endocrinology?

- Thyroid, neck near larynx, rate of metabolism
- Testes, lower pelvic area, produces androgen, initiates secondary male characteristics
- Ovaries, pelvic area, produces oestrogen, initiates secondary female characteristics
- Pancreas, close to stomach, produces insulin, regulates sugar levels
- Pituitary, base of brain, regulates insulin production, blood pressure, skeletal growth
- Adrenal, above the kidneys, produces adrenalin

8 – Hormones – aldosterone

- 1 Blood or sap (in plants) 2 Feedback
3 Carbon, hydrogen and oxygen
4 Kidney
5 Charged particles, allow bonding to occur with other charged ions
6 Hypo = lower/less, hyper = higher/greater
7 Blood test, that's where hormones are found
8 System of glands that release hormones into circulatory system
9 **a** Over a long period of time **b** release from e.g. gland **c** signs of disease **d** manufactured **e** watched closely
10 Low urine output, high blood

- pressure
11 Reduces inflammation
12 Low red blood cell count, reduces the amount of oxygen that can be delivered to tissues and organs
13 Brain, *sends message to*, Endocrine system, *changes level of hormone production*, Altered level of hormone in blood, *blood reaches organ*, Targeted organ responds, *signal sent to brain*

9 – Nerves

- 1 Teacher to check, e.g. an elastic bundle of tissue/fibres that transmit electrical impulses from the brain, etc.
2 Teacher to check
3 **a** Conducts impulse away from nerve body **b** Myelin sheath **c** Controls cell **d** Dendrite **e** Gap between nerve cells **f** Neurotransmitter
4 Brain and spinal cord
5 Half a sphere
6 Nerve cell
7 Rounded projection
8 Cerebellum, cerebrum and medulla
9 Increased surface area
10 Brain detects nerve activity, sends instructions via the spiral cord to the relevant sites
11 Teacher to check, e.g. nerves are associated with quick reactions, nervous people are twitchy

10 – Health issues – the brain

- 1 Teacher to check, e.g. complicated organ, not easy to see 'what does what'
2 Teacher to check; e.g. recognise faces, dangers, recognise a voice on phone, etc.
3 They are addictive and harmful to the health
4 Teacher to check
5 Impulses, one, dendrites, axon, many, long, end
6 Top row: neurons, information, 18 years. Bottom row: dendrites, cell body, axons
7 Toxic to the nervous system
8 Refers to the senses, e.g. touch, sight
9 Teacher to check; e.g. Cowper's gland, Calvin cycle

11 – Human reproduction – female

- 1 Fraternal
2 Each received slightly different genetic material from their parents
3 About 16 cm long, 2 cm thick, 600 g
4 Foetus leans on partially full bladder
5 **a** Teacher to check; e.g. releases sex cell at regular intervals automatically

- b** Muscles contract to help push the baby along the canal
6 **a** Fimbriae, 'Catch' eggs and draw them into oviduct. **b** Ovary, houses ova/eggs **c** Mouth of uterus **d** Cervix, upper birth canal **e** Vagina, lower birth canal, organ of intercourse **f** Vulva, opening **g** Urethra, passage of urine **h** Endometrium, uterine lining, shed **i** Uterus, muscular, houses embryo **j** oviduct, path for ova
7 **k** Fallopian tube, conducts eggs from ovary to uterus. **l** Ovary, houses eggs **m** Uterus, chamber where embryo grows **n** Cervix, upper birth canal **o** Vagina, lower birth canal
8 Teacher to check
9 Approximately half way along Fallopian tube
10 Approximately halfway down uterus, on lining

12 – Human reproduction – male

- 1 Voice deepens, facial hair appears, testes drop, pubic hair grows
2 Sex cell
3 Deoxyribonucleic acid
4 Sperm, food, fluids
5 Biological catalyst
6 Scrotum
7 Chemical produced by a gland that has an affect elsewhere in the body
8 Testosterone – helps sperm mature, allows puberty to begin
9 Produce energy through cellular respiration
10 Puberty
11 Bent tail and two tails cannot swim properly, two heads may have double the DNA or double of some genes, micro has too little DNA, macro may have too much
12 **a** Midpiece, contains mitochondria to provide energy for tail. **b** Tail, propels sperm forward. **c** Cap, 'acrosome' contains enzymes to digest outer coat of oocyte. **d** Head contains DNA
13 Able to move
14 Kills sperm
15 **a** True **b** False **c** True **d** False **e** False **f** True **g** True **h** False
16 **e** Seminal vesicles, lubricating and nourishing fluid. **f** Prostate gland, alkaline fluid to coat sperm. **g** Cowper's gland, nourishing fluid for sperm. **h** Epididymus, sperm matures here. **i** Testis, sperm forms in seminiferous tubules. **j** Scrotum, muscular sac, houses testis. **k** Seminiferous tubule. **l** Urethra
17 Teacher to check



Selected answers (cont'd)



13 – Human reproduction crossword

Down: 1 – cowers; 2 – embryo; 5 – puberty; 6 – ovum; 7 – enzymes; 8 – seminiferous; 12 – prostate; 13 – vagina; 15 – fallopian tubes; 17 – urethra; 18 – testosterone; 19 – testes; 23 – fats

Across: 3 – womb; 4 – vulva; 9 – spermatazoa; 10 – endometrium; 11 – ovary; 14 – bones; 16 – oviduct; 20 – semen; 21 – hormone; 22 – placenta; 23 – foetus; 24 – mitochondria; 25 – alkaline

14 – Human reproduction quiz

- 1 Nourishes sperm, provides protective coating against acidic nature of vaginal fluids, provides lubrication
- 2 Puberty is the onset of secondary sex characteristics
- 3 To protect it against infection, etc.
- 4 It needs them for healthy growth and development
- 5 Both are sex hormones that trigger onset of puberty and secondary sex characteristics. Testosterone is the main male hormone, oestrogen the main female one
- 6 In males it is used to pass urine and semen from the body, in females it is only used to pass urine
- 7 An embryo describes the developing baby from conception/ 1 days (depending on belief) to 8 weeks, a foetus is the baby from 8 weeks to birth
- 8 Sperm is designed solely to deliver DNA, egg is designed to carry its DNA and to nourish the early weeks of the fertilised egg

15 – Ethics – stem cell research

- 1 Teacher to check
- 2 Stem cell
- 3 Flexible structure connecting the developing baby to the placenta, through which nourishment passes to the baby and waste passes away. Baby moves to eating, breathing and passing its own waste once it's born
- 4 May wish to help others, personal feelings, religious beliefs
- 5 Teacher to check. Many degenerative diseases
- 6 A person who has all of their arms and legs paralysed
- 7 Less than 8 weeks after conception
- 8 More than 8 weeks after conception

16 – Plants

- 1 Root system
- 2 Stem system
- 3 Yes, it fits description

4 To be checked, e.g. any flowering bulb, sweet potato

5 Spreading seeds, rhizomes

6 Yes

7 **a** True **b** True **c** False **d** True **e** False **f** True **g** True

8 Teacher to check, any non-woody plant

9 Teacher to check. Any plant with woody bark

10 Xylem transports raw materials and water up from the roots. This is used to manufacture food. Phloem then transports the food throughout the plant

11 Any moist tropical or sub-tropical region. They 'like' those conditions

12 No, they don't perform the functions of a true leaf

13 They cover rocks, etc. until larger plants can take over

14 They're so tiny and delicate

15 They lack a vascular system

16 Algae used as food by many marine species, used to hide in, lay eggs, etc. Grasses feed herbivores

17 Teacher to check, non-bryophyte species

18 Fern

19 Teacher to check, any flowering plants

20 Softwood, the seeds themselves are not covered by a protective coating once they are in a position to germinate

21 Pollen

22 Ovule

23 Grow new plants, food, seasoning and preserve species

17 – Transpiration

- 1 Gas, steam, etc.
- 2 Yes, it involves losing water vapour through pores
- 3 Example veins and arteries; blood, lymph
- 4 Teacher to check. Transpiration is the release of water vapour from pores in the plant. Evaporation occurs when water heats up and is converted to vapour
- 5 Get food from eating other organisms
- 6 Yellow, any naturally yellow plant
- 7 Answers columns left to right Food, photosynthesis, any two from chloroplasts, cytoplasm, nucleus, cell wall, chlorophyll, cell membrane, phloem, food, xylem, water.
- 8 For water to enter stalk and regain stiffness

18 – Transpiration quiz

1 B 2 C 3 D 4 A 5 B 6 B

7 True

8 True

9 False

10 True

11 False

12 False

13 Answers found in text provided

14 Increase surface area of roots

15 Cohesion; attraction between same type of molecules, adhesion; attraction between unlike molecules

19 – Transpiration – graphing

- 1 Teacher to check
- 2 Teacher to check
- 3 Transpiration maximum during the day, minimum at night
- 4 Lower transpiration rate
- 5 e.g. Average per hour x 24 hours
- 6 $71/51 = 1.39 \text{ L/hr}$, $24 \times 1.39 = 33.4 \text{ L}$

20 – Soil treatment and seedling growth

- 1 Best, most favourable
- 2 Calcium
- 3 Something that is kept the same, other factors are compared against this
- 4 Same type of seed, same type of soil, same water, heat, etc.; i.e. only change one variable
- 5 13.1 cm
- 6 19.7
- 7 14.0 g
- 8 Height: 21.3, 43, No of leaves: 46, 108, Mass 134, 252.
- 9 **a** True **b** False
- 10 Treat soil with lime, choose different plant
- 11 pH 4.5 to 7, blue, above pH 7, pink. White flowers not affected

21 – Germination of Australian native species

- 1 **a** Young plants grown from seed
- b** For an extended period of time
- c** Begin to grow
- d** Biological activities inactive
- e** To hold in check
- f** Copy
- g** Place where an organism lives
- h** Rubbed away
- i** Group of similar organisms that can interbreed
- j** Putting into a liquid
- k** Allowed to come into contact with
- l** Dissolve out soluble substances by slowly passing water through



Selected answers (cont'd)



- 2 Teacher to check. Most hard-coated seeds need pre-treatment; e.g. acacia, can use boiling water, smoke, heat. Some need cold period, e.g. Snow gum
- 3 Teacher to check, e.g. recycle grey water, plant drought-resistant plants.

22 – Cloning and tissue culture of plants

- 1 **a** Children, progeny **b** Many cells
c Living things **d** No longer existing
e Very small **f** Typical samples
g Capable of surviving **h** Group of similar cells **i** Grown under artificial conditions **j** Substances that provide nourishment/food **k** Reproduce giving identical cells **l** Genetically identical
- 2 e.g. Advantage is the species can exist where there are few or no others like it. Can 'buy time' until others arrive. Disadvantage is loss of genetic variation
- 3 Cloning produces a genetic twin. Inbreeding leads to loss of genetic variation but some can still be found
- 4 In order: B, D, F, C, G, A, E, H

23 – Seeds in deep freeze for disaster

- 1 **a** Association **b** Permanently frozen layer of ground **c** Sea wave caused by e.g. earthquake
d Underground chamber
e Scattered islands **f** Climate change – heating up of average Earth temperature **g** Resource
h Cultivation and farming
i Store **j** Range **k** Pattern of weather
- 2 $50 \times 50 \times 5 = 12\,500 \text{ m}^3$ (cubic metres)
- 3 Unreactive, inexpensive, easy to shape, seals well
- 4 Teacher to check, e.g. to preserve genetic diversity
- 5 Secure establishment where USA stores its gold reserve
- 6 Should extend beyond the idea that so long as we have seed stores we're safe. Looking at the bigger picture so that they see just saving things
- 7 Students should think of things like the time it will take for the crops to grow and be harvested, what will we eat until then? What will the cost of these crops be until we can re-establish crops on a larger scale. Students to think of the fact that if the people themselves aren't suffering from radiation poisoning, the land and water probably would be
- 8 Could argue that water is more important. After all, there are people with purely fish diets etc. but without water, we'd all be dead. Without it, the seeds wouldn't survive either

- 9 Teacher to check, should have good reasons
- 10 Teacher to check for logic; students should mention seeds, planting, machinery, labour, water, raw materials, distribution, etc.

24 – Biodiversity and conservation

- 1a Different species of organism in a particular area
b Area in which an organism lives
c Bionetwork, interaction of organisms and their surroundings
d Organisms that can interbreed and produce fertile offspring
e Vegetation **f** Harvest
g No organisms of this type still living
h Preserve and protect
i In risk of dying out
j Capable of succeeding, e.g. germinating
- 2 More money, can concentrate on one crop, disease could wipe out one crop, markets could change. 'Eggs in one basket'

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